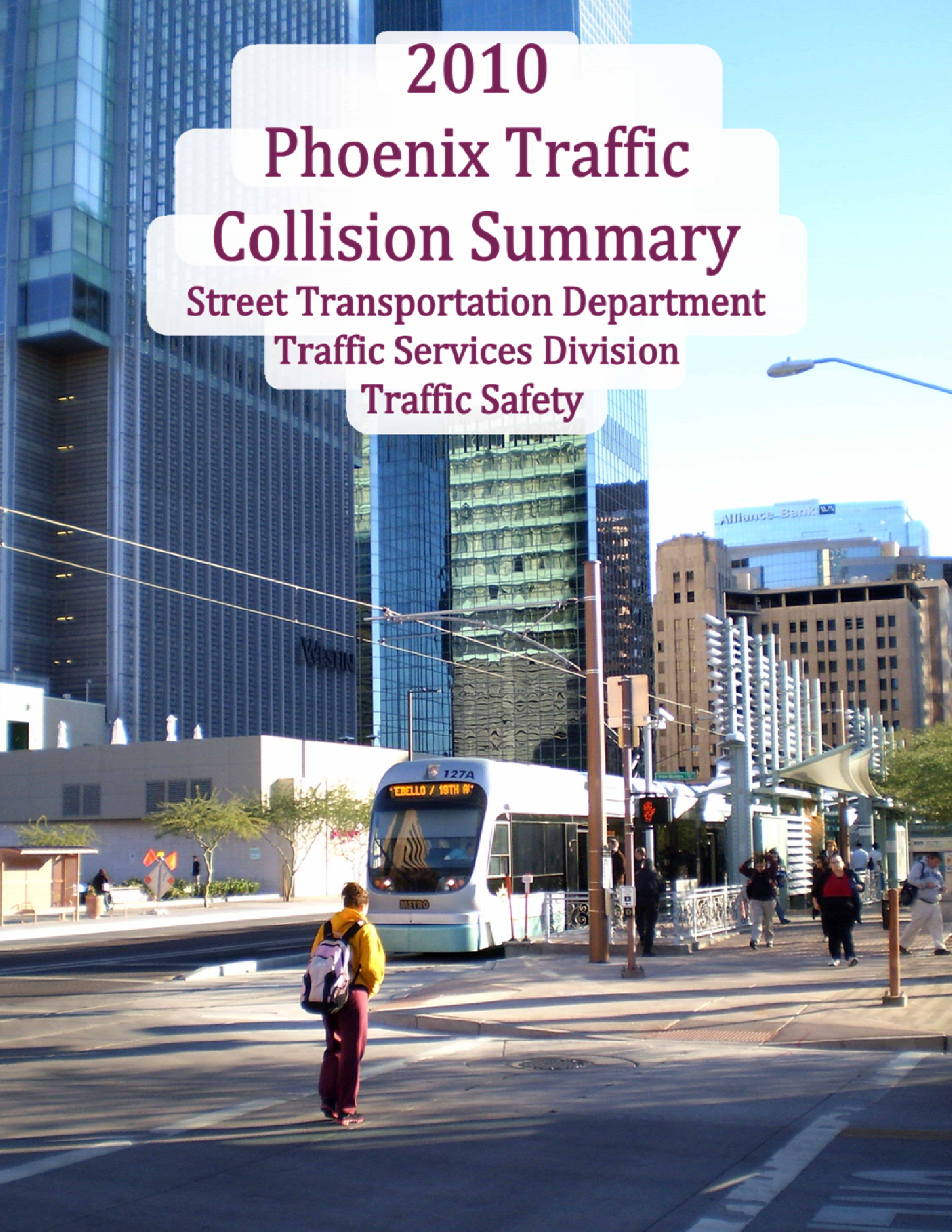
2010 Phoenix Traffic Collision Summary

Street Transportation Department
Traffic Services Division
Traffic Safety



2010 Phoenix Traffic Collision Summary
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Introduction

Traveling is complicated. Each foot, yard or mile traveled requires hundreds or thousands of individual decisions that are made so effortlessly by the human mind that we are seldom aware that they are being made, yet one wrong or poorly timed decision can end in disaster. When viewed in light of the fact that each fellow traveler is making a similar series of decisions with similar consequences, it is easy to understand how complicated and even dangerous travel, whether by car, bike or foot, can be. Yet the fact that the overwhelming majority travel ends with people arriving at their destinations safely is testament to not only how safe our travel environment is but how well people operate within this environment. Despite this relative safety, traffic crashes do still occur and people are sometimes injured or even killed as a result.

Looking back on 2010, Phoenix saw another record low number of traffic crashes. The 21,963 crashes reported in 2010, are not only less than 2009 crash totals but represent the lowest number of crashes in over 20 years. The Phoenix numbers also mirror decreases state and national crash numbers and although traffic crashes are tied closely with the economy, the decreases have been encouraging.

The number of crashes that resulted in injuries also declined in 2010 along with the total number of people injured which dropped from 12,318 in 2009 to 12,143 in 2010. There were also declines in fatal collisions and in the number of people killed in those collisions in 2010 with 108 fatal collisions and 121 fatalities. Pedestrians accounted for about 1/3 of these fatalities with 45 pedestrians killed during 2010 which is an increase from pedestrian fatalities in 2009. Unfortunately, crashes involving bicycles were close to average in 2010 as were the number of bicyclists injured and killed during the year but crashes involving motorcycles continued a decline that started in 2006 dropping to 495 in 2010 from 627 in 2006 with much more dramatic declines in the number of injuries and the number of fatalities from these crashes. Crashes involving METRO light rail which saw its second complete year of operations in 2010 declined from 35 in 2009 to just 18 in 2010.

From an enforcement standpoint, the percentages of run red collisions in 2010 remained the same as in 2009 at just under eight percent of all crashes and the percentage of run stop sign crashes remained roughly average to the previous five years at just over one percent of all crashes. Alcohol related crashes declined to just under six percent of all crashes but alcohol was a factor in about 43 fatalities or about a third of all fatal crashes.

If anything good can come of an individual traffic crash, it is that safety advocates across the country and here in Phoenix look to the tragedy of individual crashes to try to make traveling safer for everyone. This Traffic Collision Summary is part of that effort.

2010 Phoenix Traffic Collision Facts-at-a-Glance

Overall

- Approximately 33 people were injured every day in 2010 as a result of a traffic crash.
- More than 2/3 of all collisions resulted in property damage only.
- More than 26 percent of all collisions occurred between the hours of 3 pm to 6 pm but only 10.7 percent of the traffic fatalities occurred during those times.
- Almost 68 percent of fatal crashes occurred during night or twilight hours.

Drivers

- In 17 percent of all crashes, the primary vehicle driver at fault was between 14 to 21 years of age. These crashes include 12 fatal and 105 serious injury crashes.
- From the above crashes, 5 fatal and 11 serious injury collisions were caused by young drivers under the influence of alcohol.
- In just under 7 percent of all crashes, the primary vehicle driver at fault was 65 years of age or older. These crashes include 6 fatal and 48 serious injury crashes.

Light Rail

- In 2010 there were 18 collisions involving a light rail vehicle. More than 77 percent of these collisions involved vehicles turning in violation of a red arrow and most of these were left turns.

Pedestrians

- 45 pedestrians died and 462 were injured in collisions with motor vehicles, which is nine more fatalities and 30 less injuries than in 2009.
- 30 percent of the collisions involved the pedestrian not using the crosswalk.

Bicyclists

- Eight bicyclists were killed and 410 were injured in collisions between bicycles and motor vehicles, which is one less fatality and 26 less injuries than in 2009.

Alcohol/Drug-Related

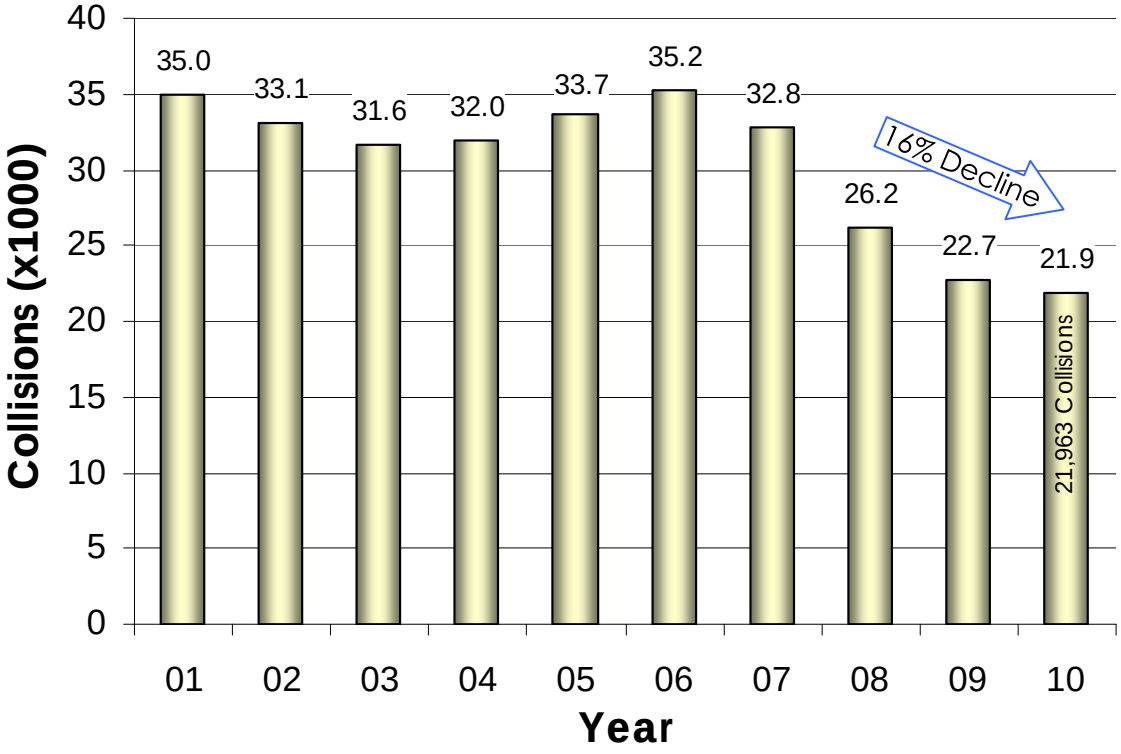
- 43 people were killed in alcohol-related collisions in 2010 which is 9 less than in 2009.
- 36 percent of the fatal crashes involved one or more drivers, pedestrians or bicyclists who had been drinking prior to the crash.
- 18 of the 20 fatal alcohol-related pedestrian collisions involved a pedestrian being under the influence of drugs or alcohol.

Red Light Running and Other Failure-to-Yield

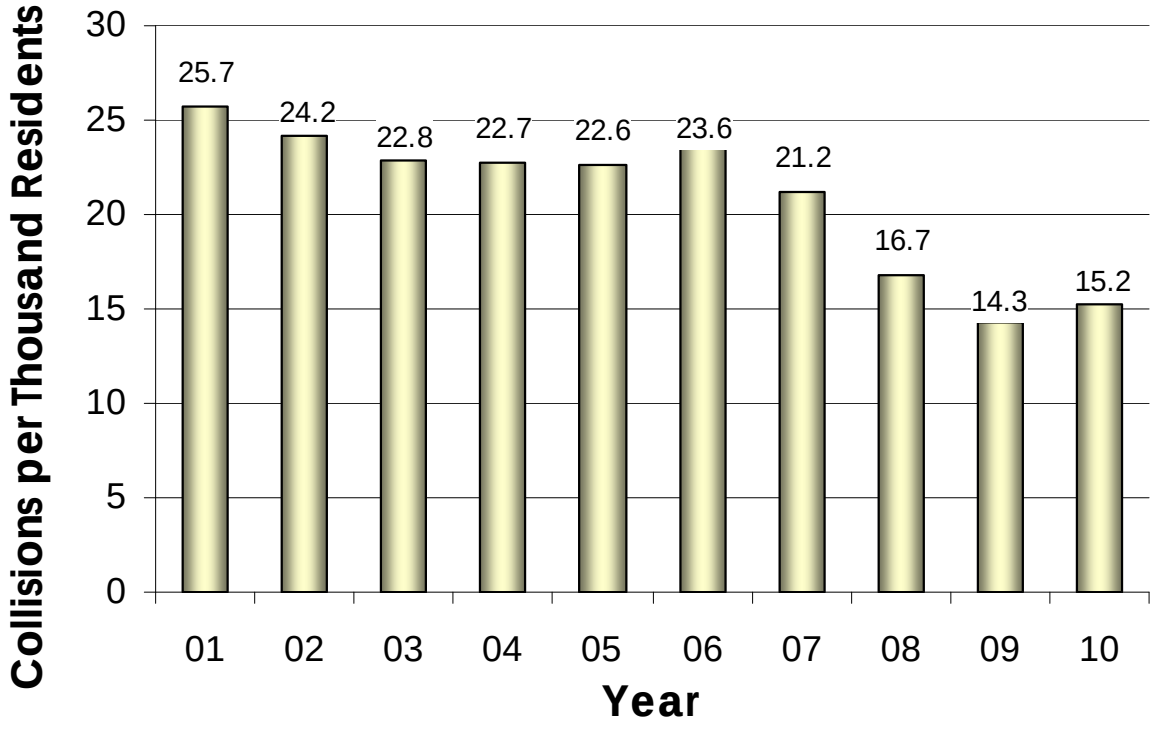
- 7 people were killed, and 1,524 were injured in collisions where at least one vehicle ran a red light 2010.
- 272 crashes resulted from vehicles running STOP signs, resulting in the death of one person and injuring 200 others.

2010 Phoenix Traffic Collision Summary

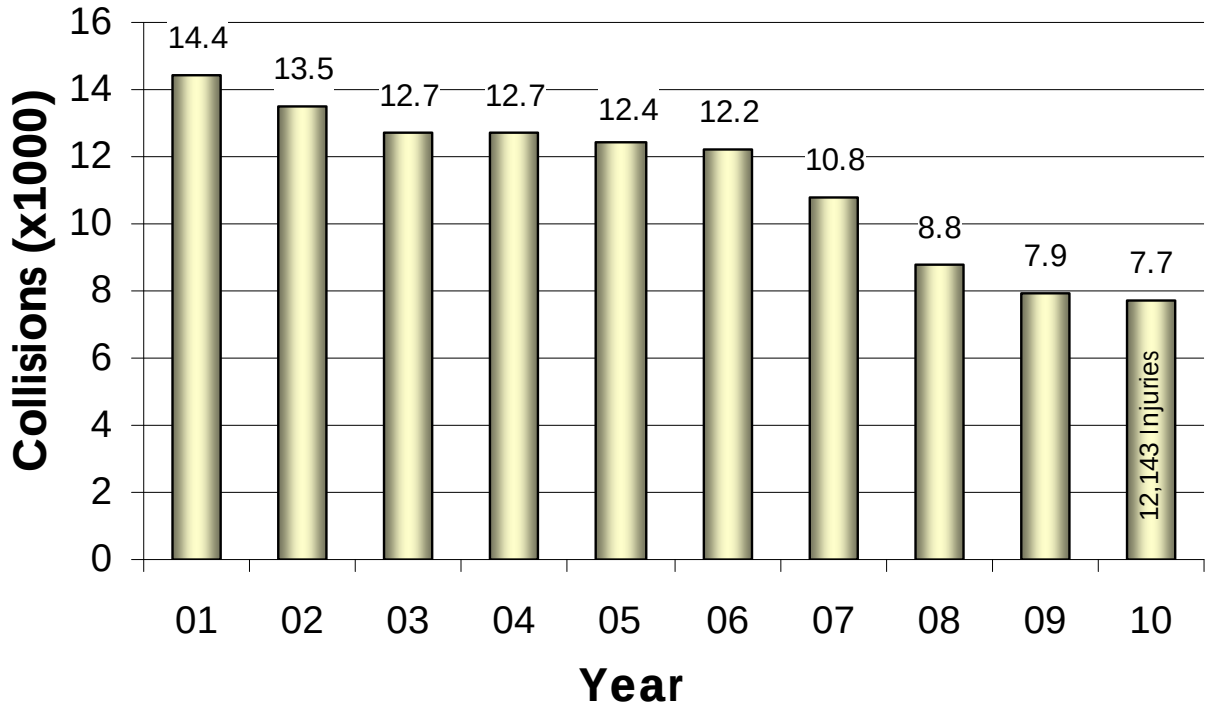
Annual Collisions 2001 - 2010



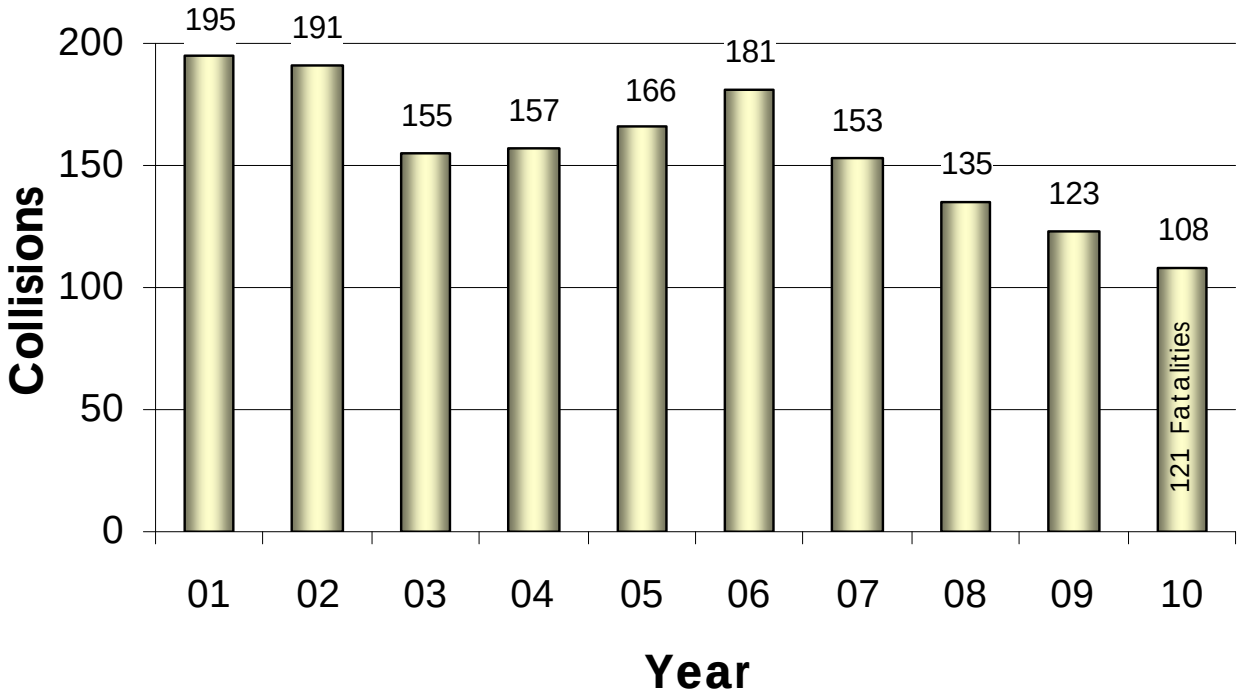
Collisions vs. Population 2001 - 2010



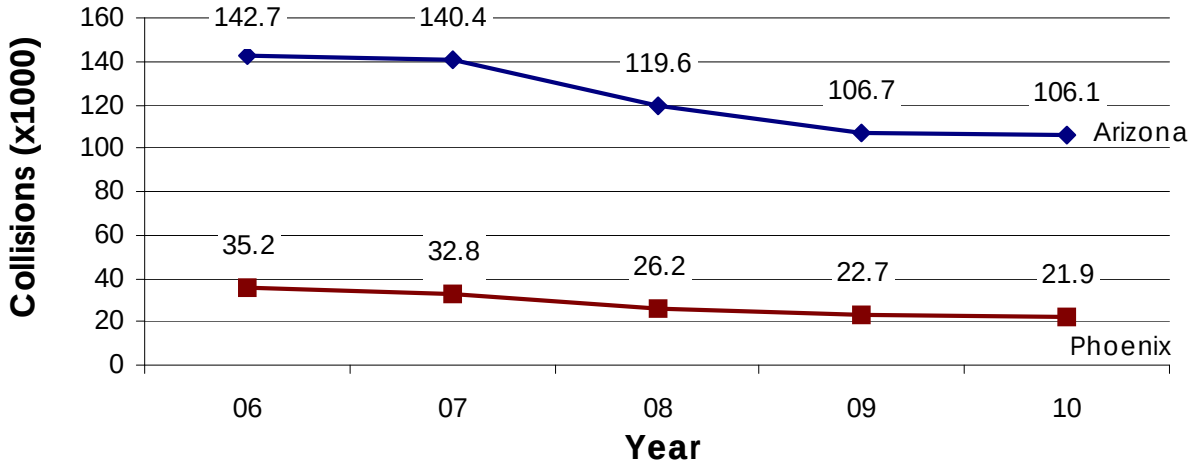
Injury Collisions 2001 - 2010



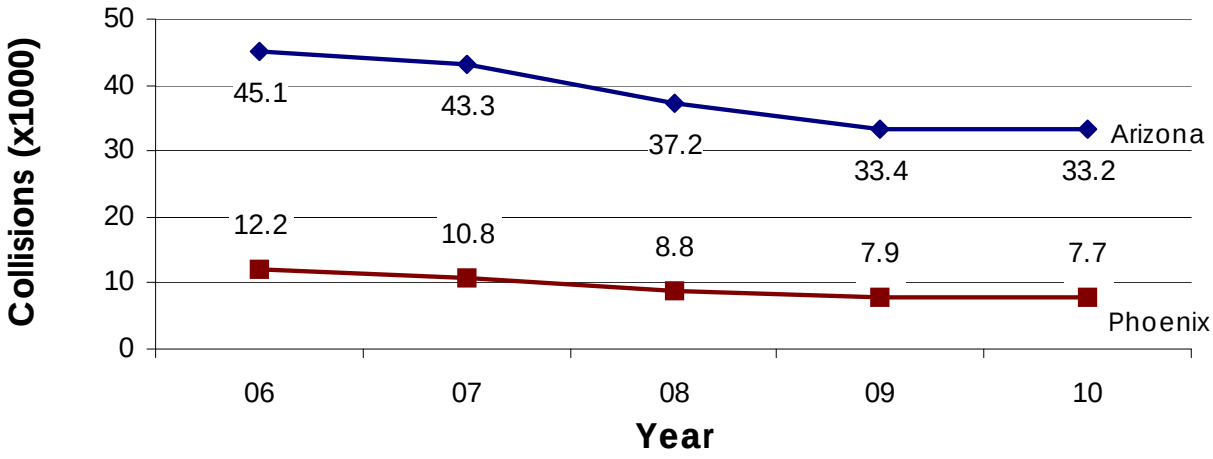
Fatal Collisions 2001 - 2010



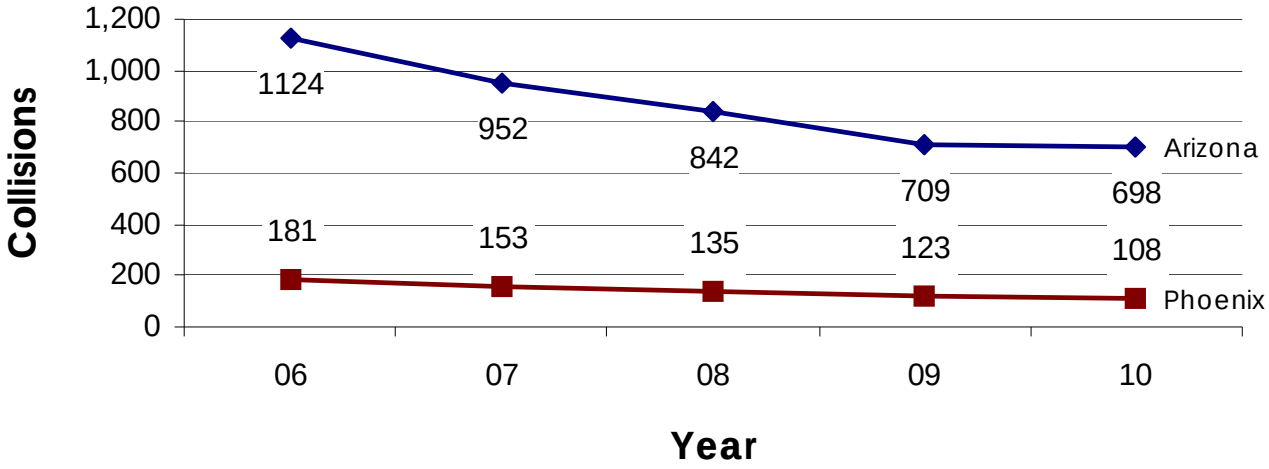
Phoenix vs. Arizona Total Collisions



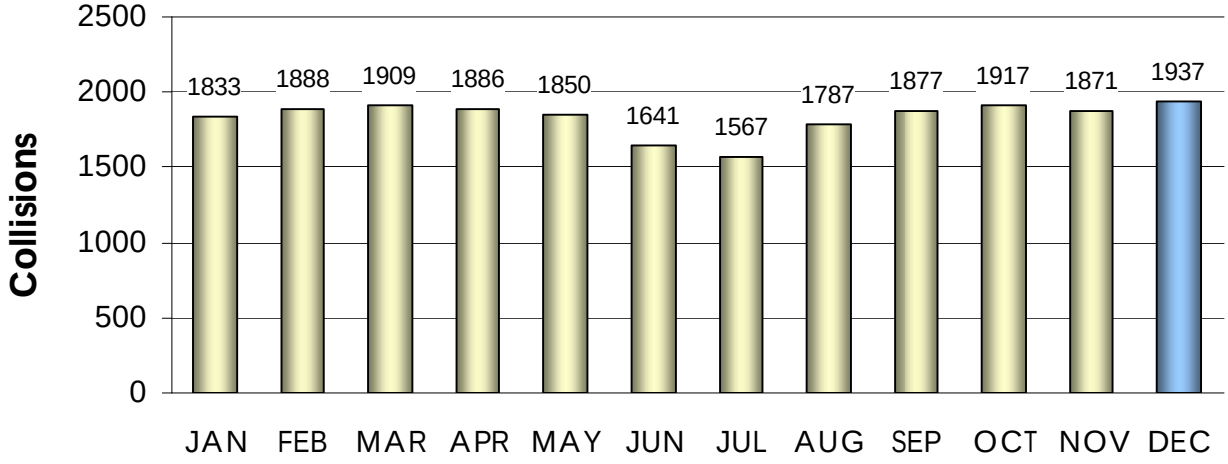
Phoenix vs. Arizona Total Injury Collisions

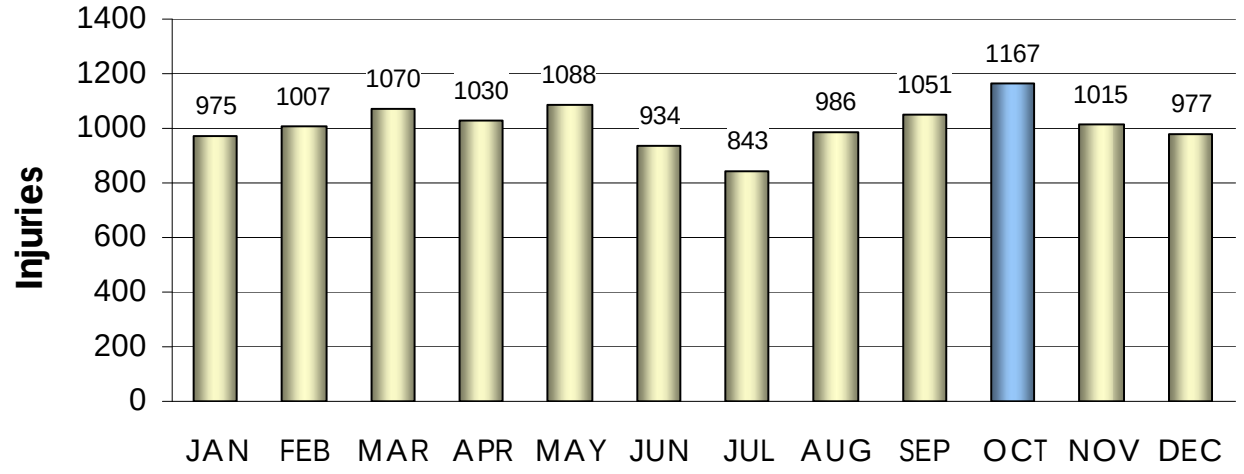


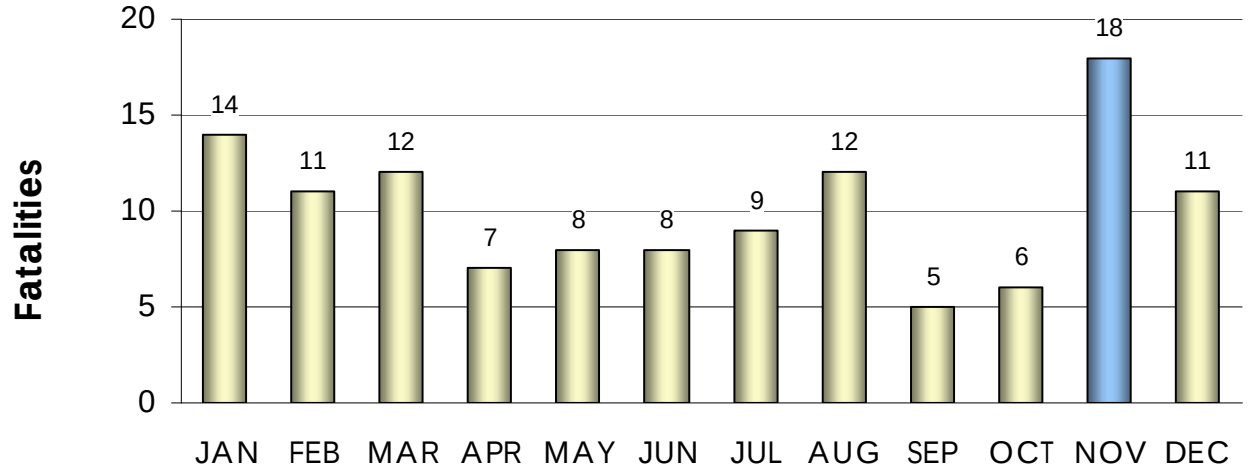
Phoenix vs. Arizona Total Fatal Collisions



Total Collisions, Injuries, and Fatalities by Month

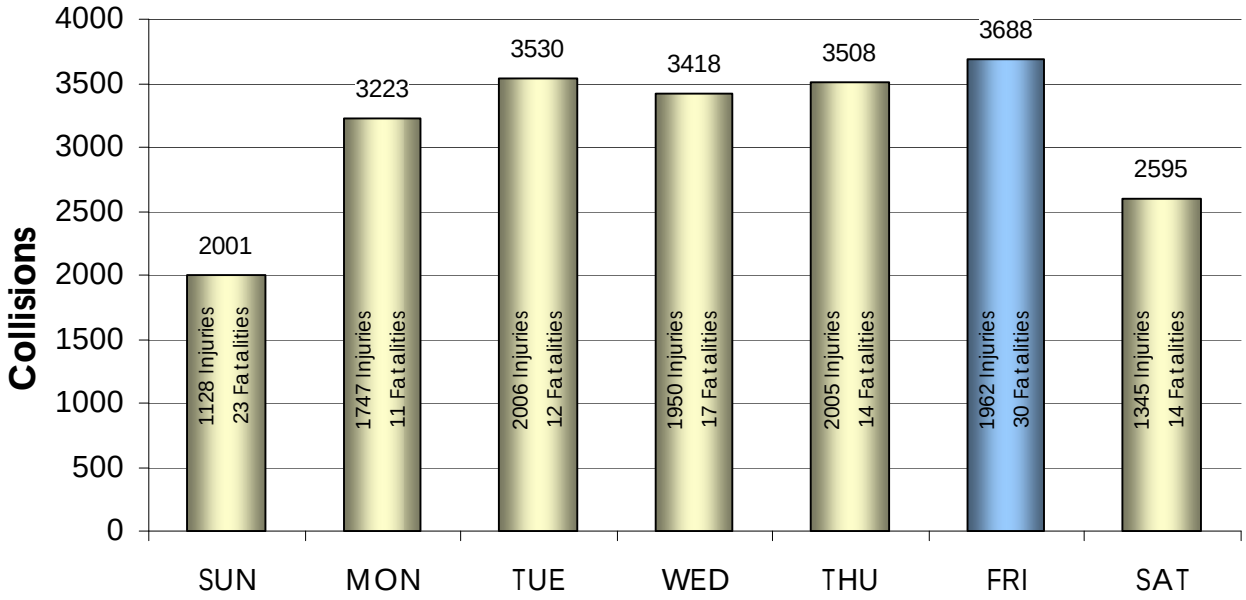




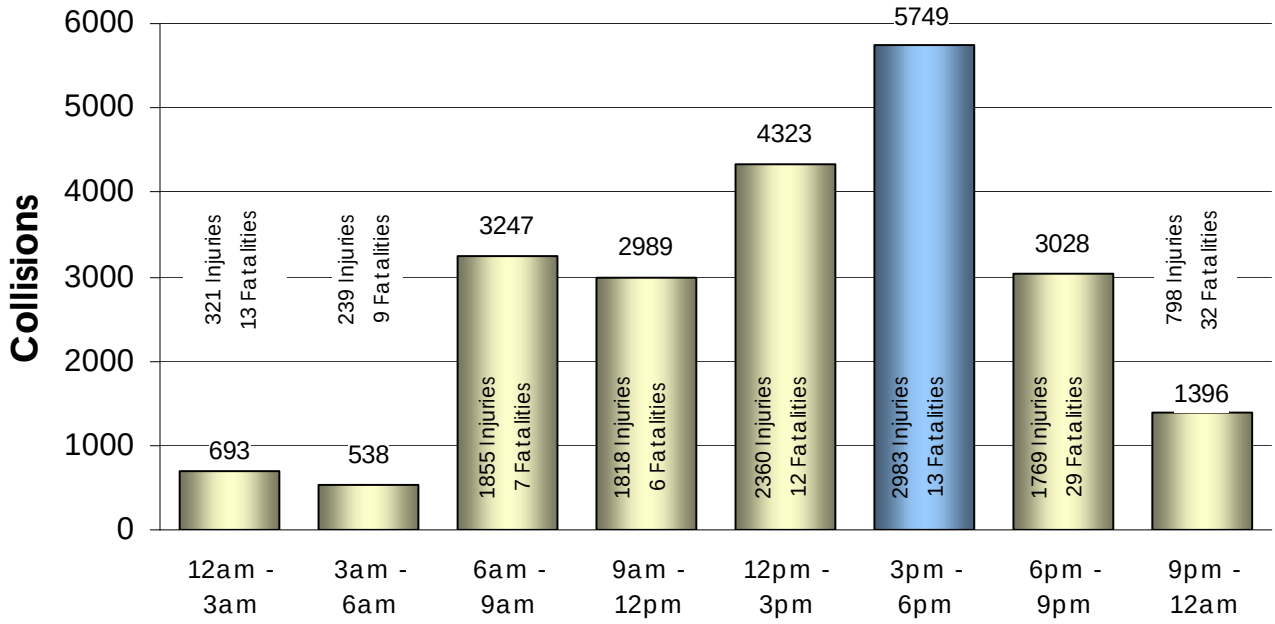


Note: Blue Column is Highest Month in 2010

Collisions by Day of Week

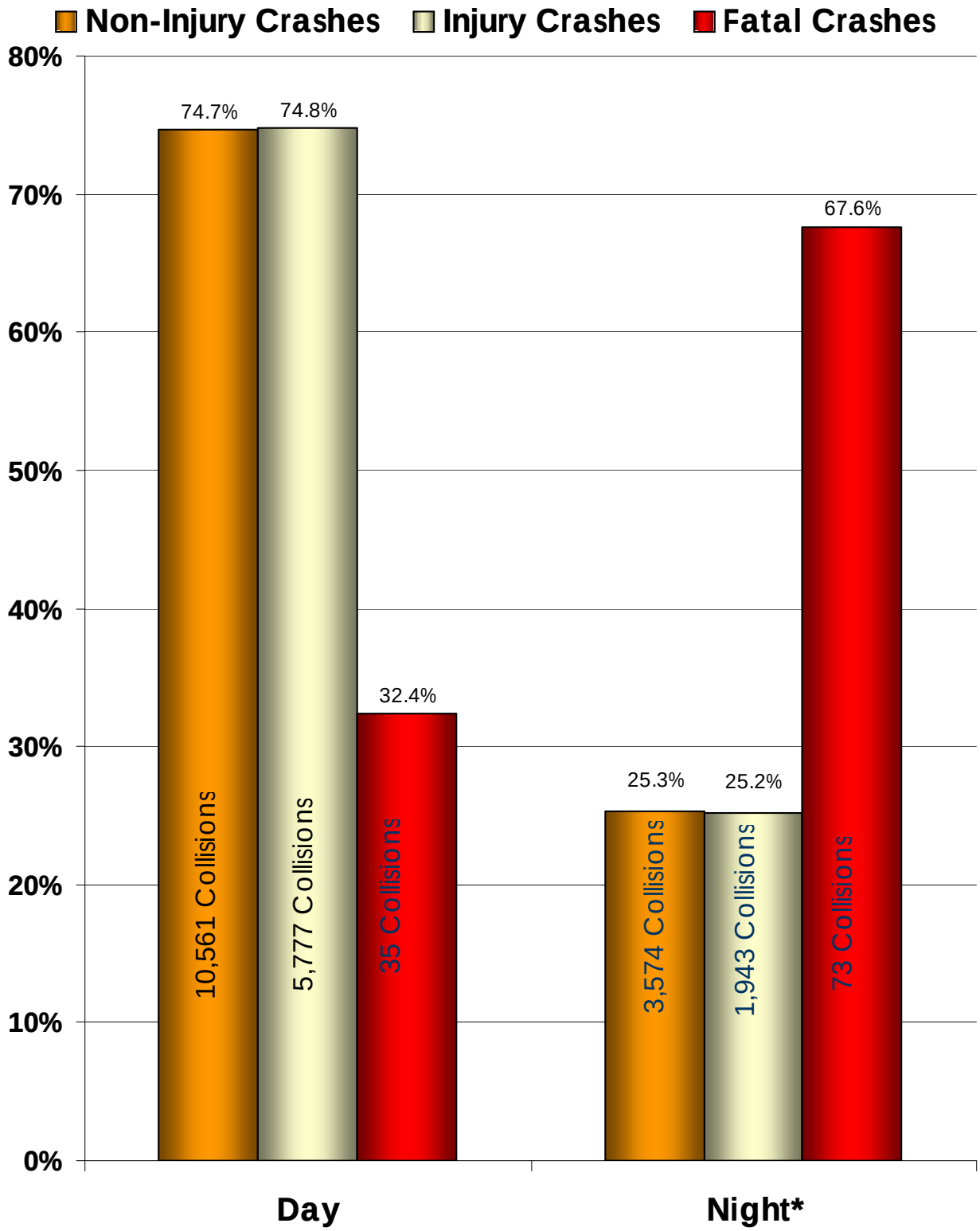


Collisions by Time of Day



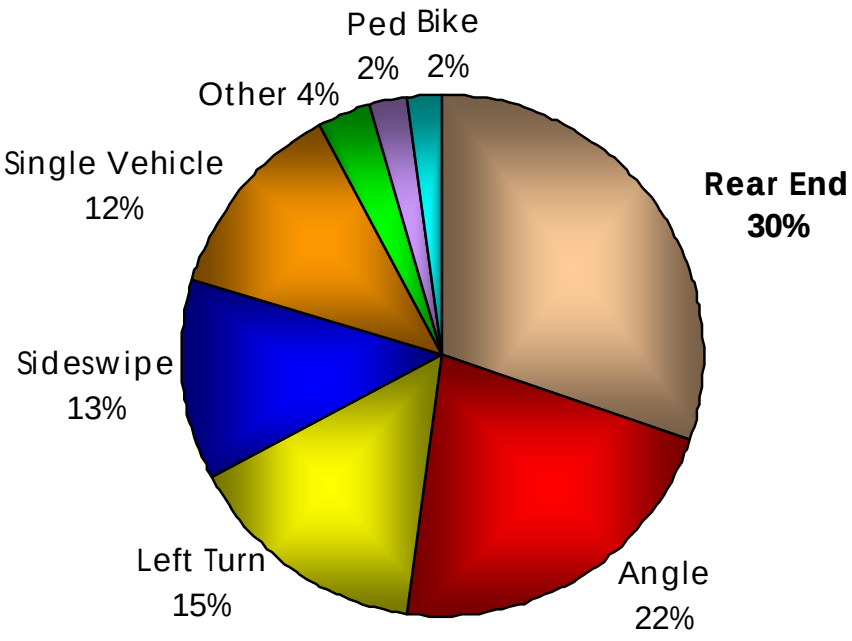
Note: Blue Column is Highest Month in 2010

Collision Severity by Light Conditions



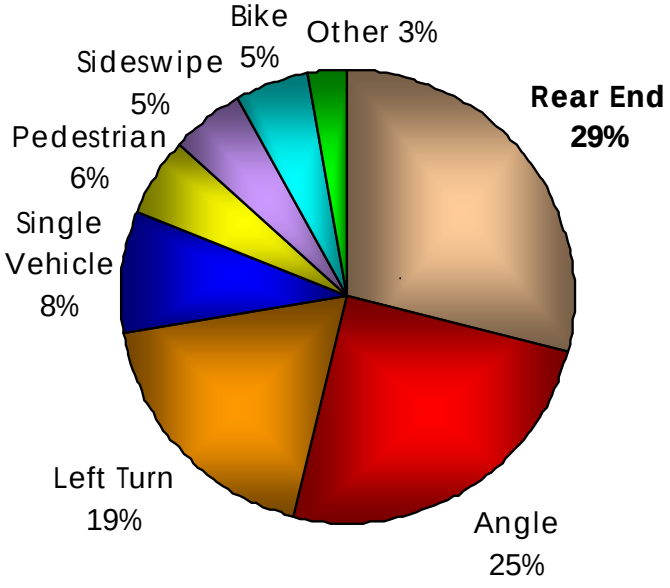
*Crashes at dawn or dusk have been combined into night crashes.

Collision Type – All Collisions

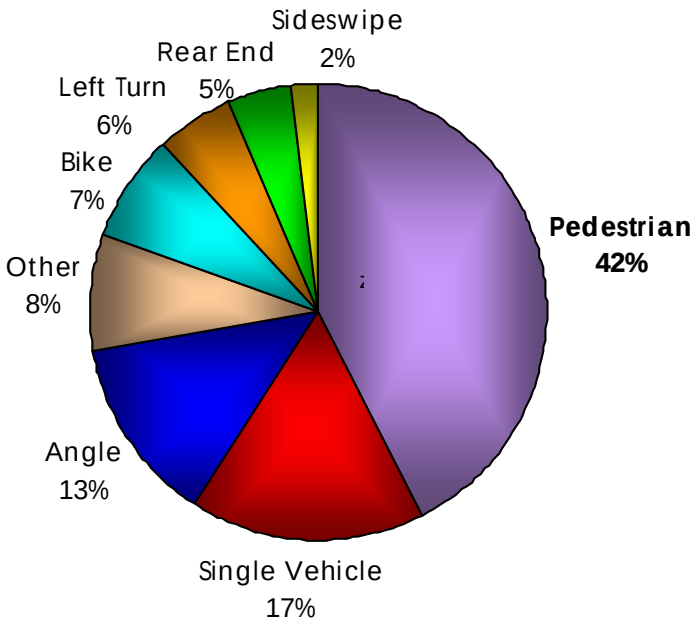


Collision Type by Severity

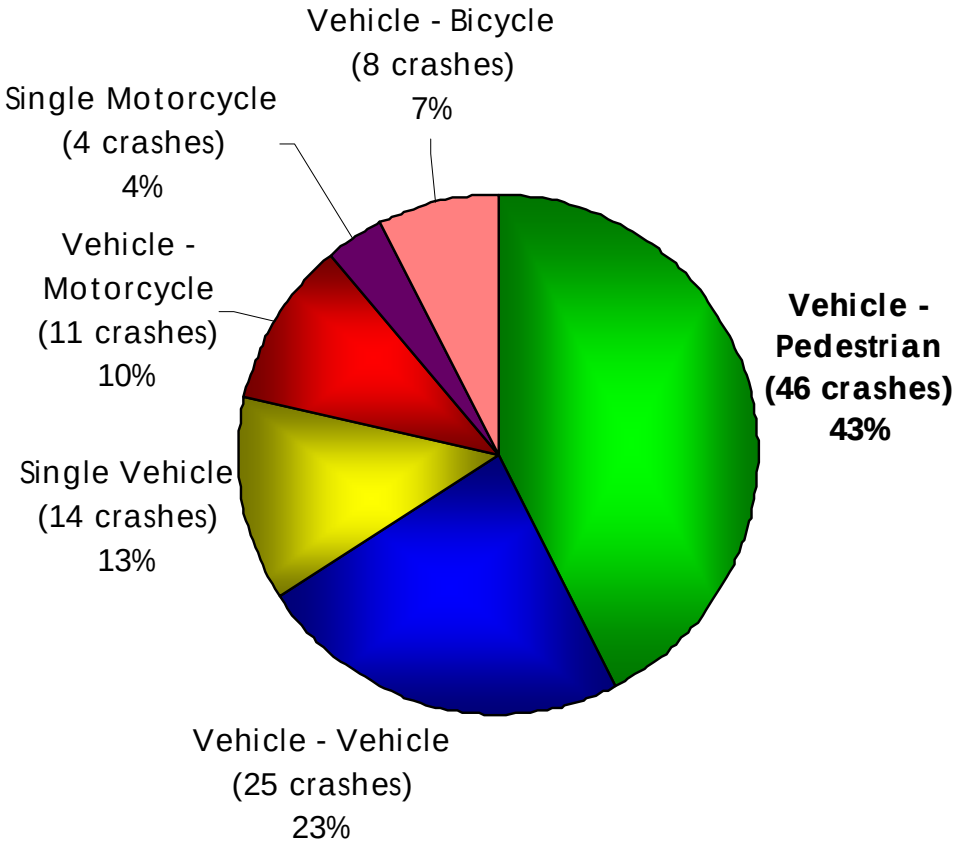
Injury Collisions



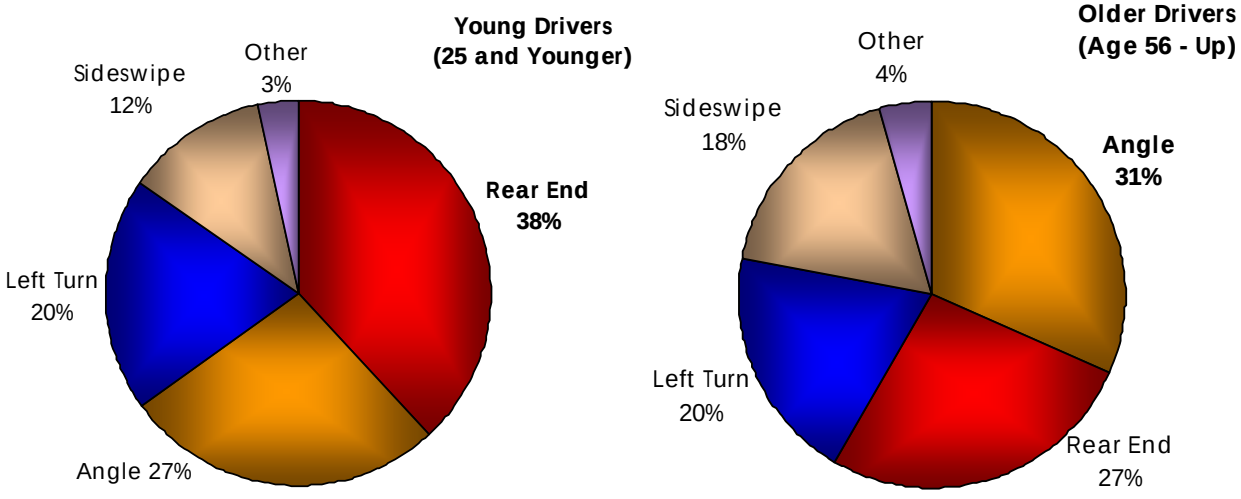
Fatal Collisions



Fatal Collisions by Mode of Transportation

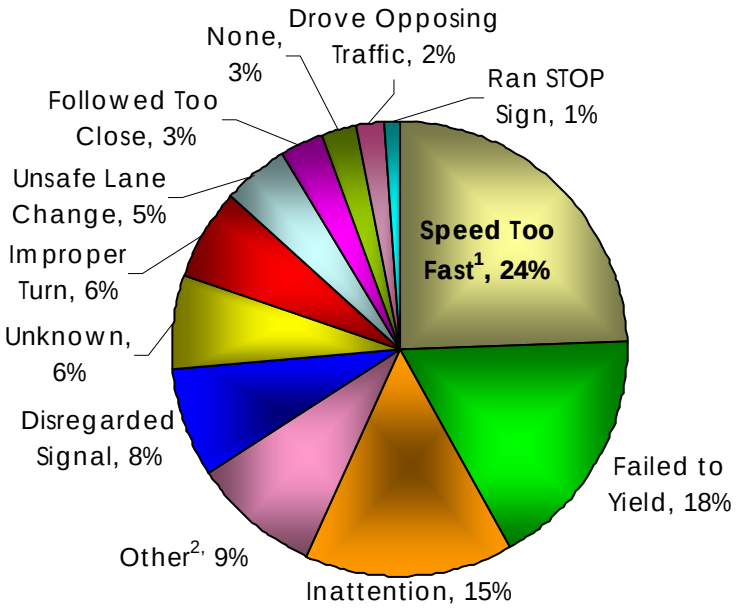


Collision Type by Driver Age Group



Only the drivers of Unit 1 vehicle vs. vehicle collisions were used for these charts.

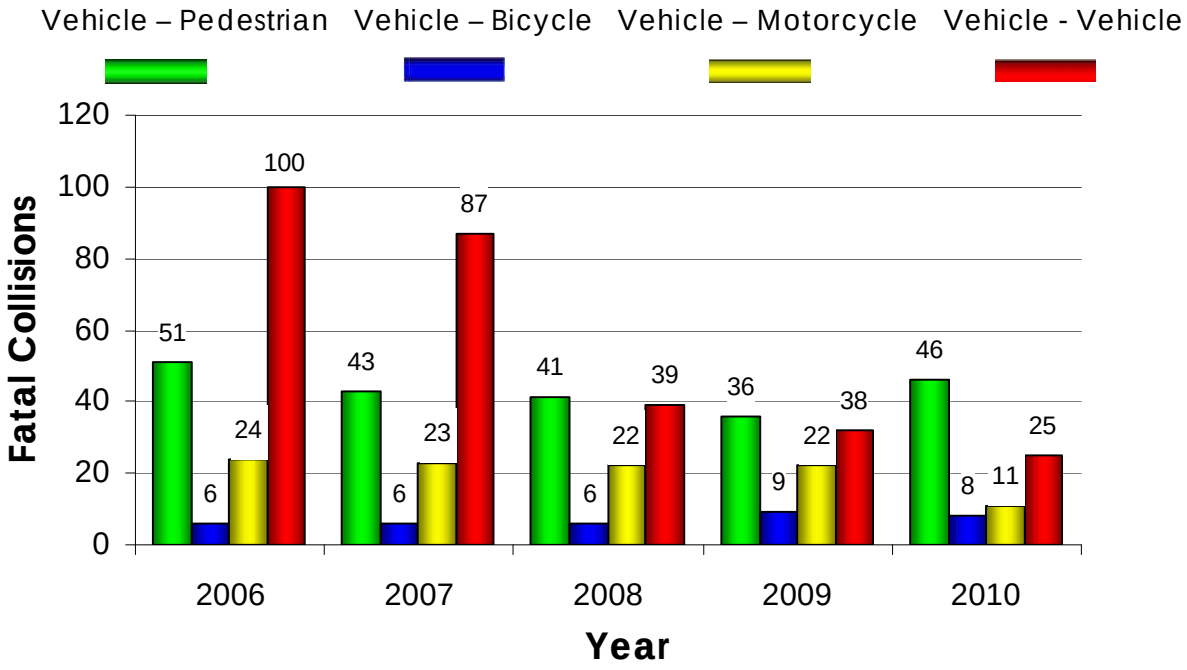
Collisions by Primary Violation



Unit 1 usually reported to be mainly at fault. Only the first violation for unit 1 was used in this chart.

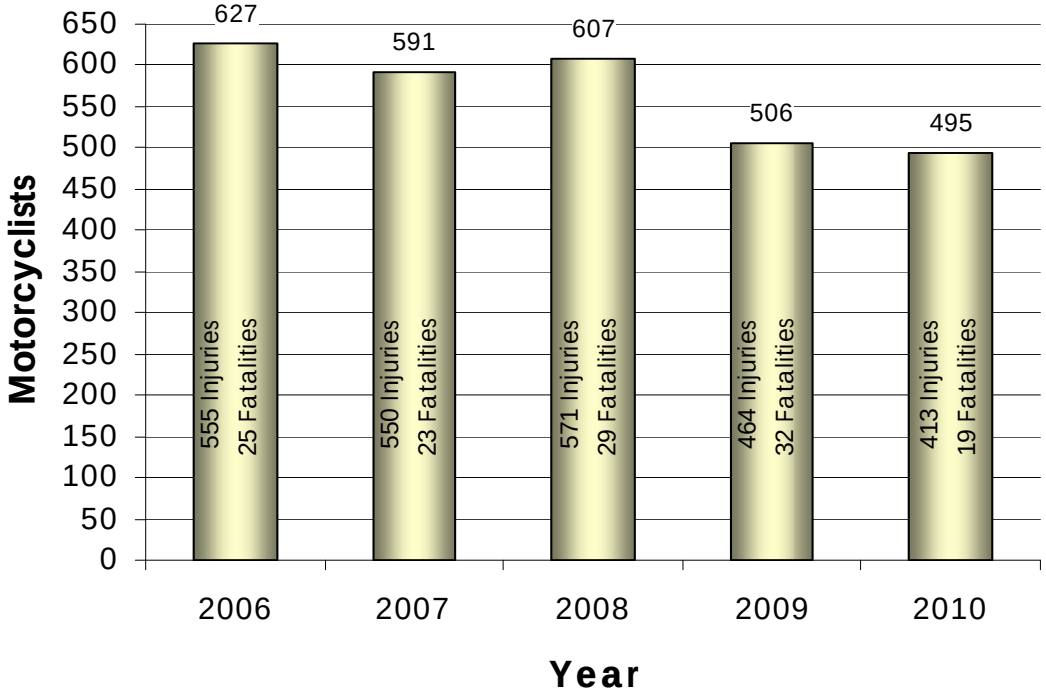
1 – "Speed Too Fast" does not necessarily mean the person was exceeding the speed limit but rather the person was traveling too fast for the conditions at the time.
2 – "Other" includes less common violation such as Unsafe Passing and Operating with Faulty Equipment.
3 – "None" refers to drivers involved in a crash without committing a violation. These instances are usually the result of uncontrollable forces such as falling debris or animals in the roadway.

Fatal Collisions by Mode 2006 – 2010*

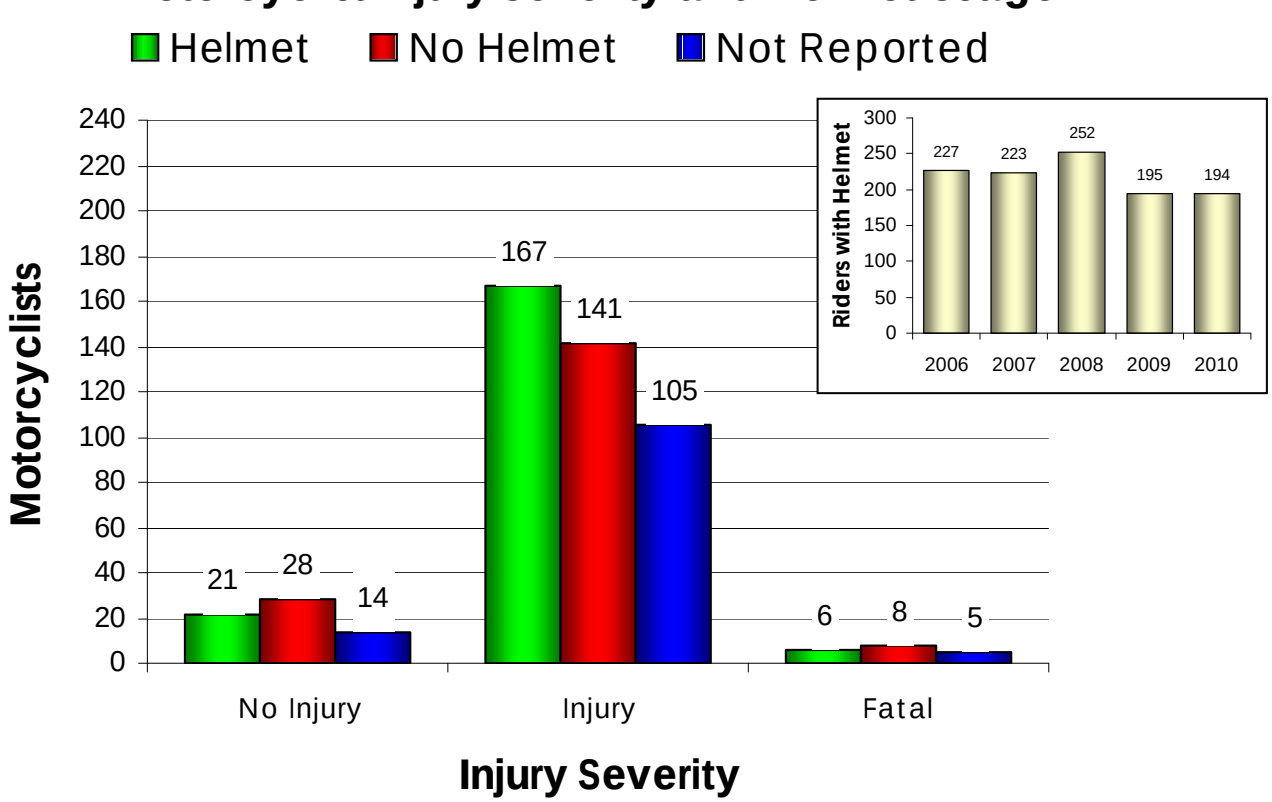


*The graph does not include single-vehicle fatal collisions

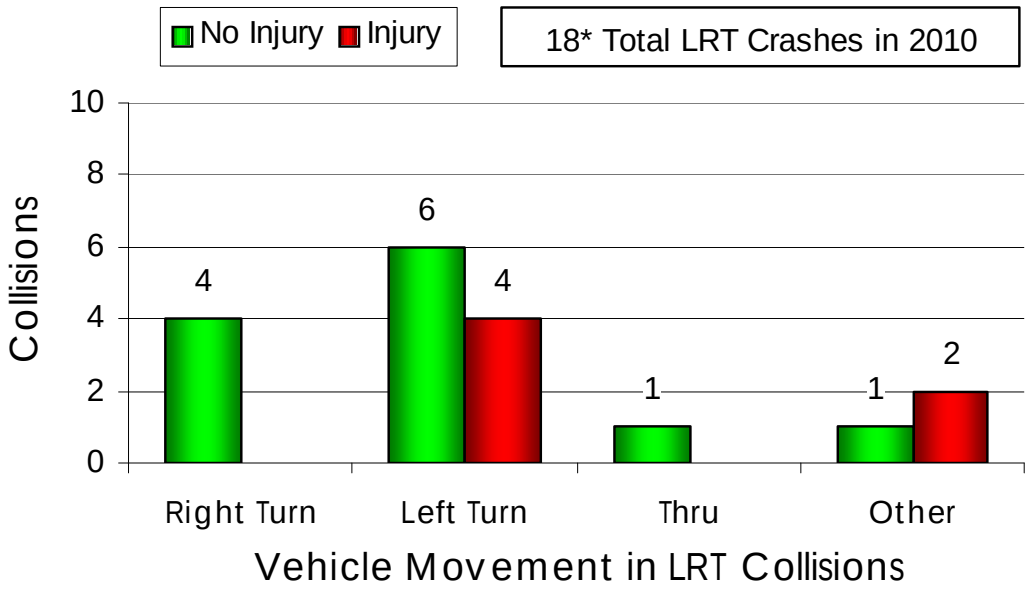
Total Motorcyclists, Injuries, and Fatalities 2006 - 2010

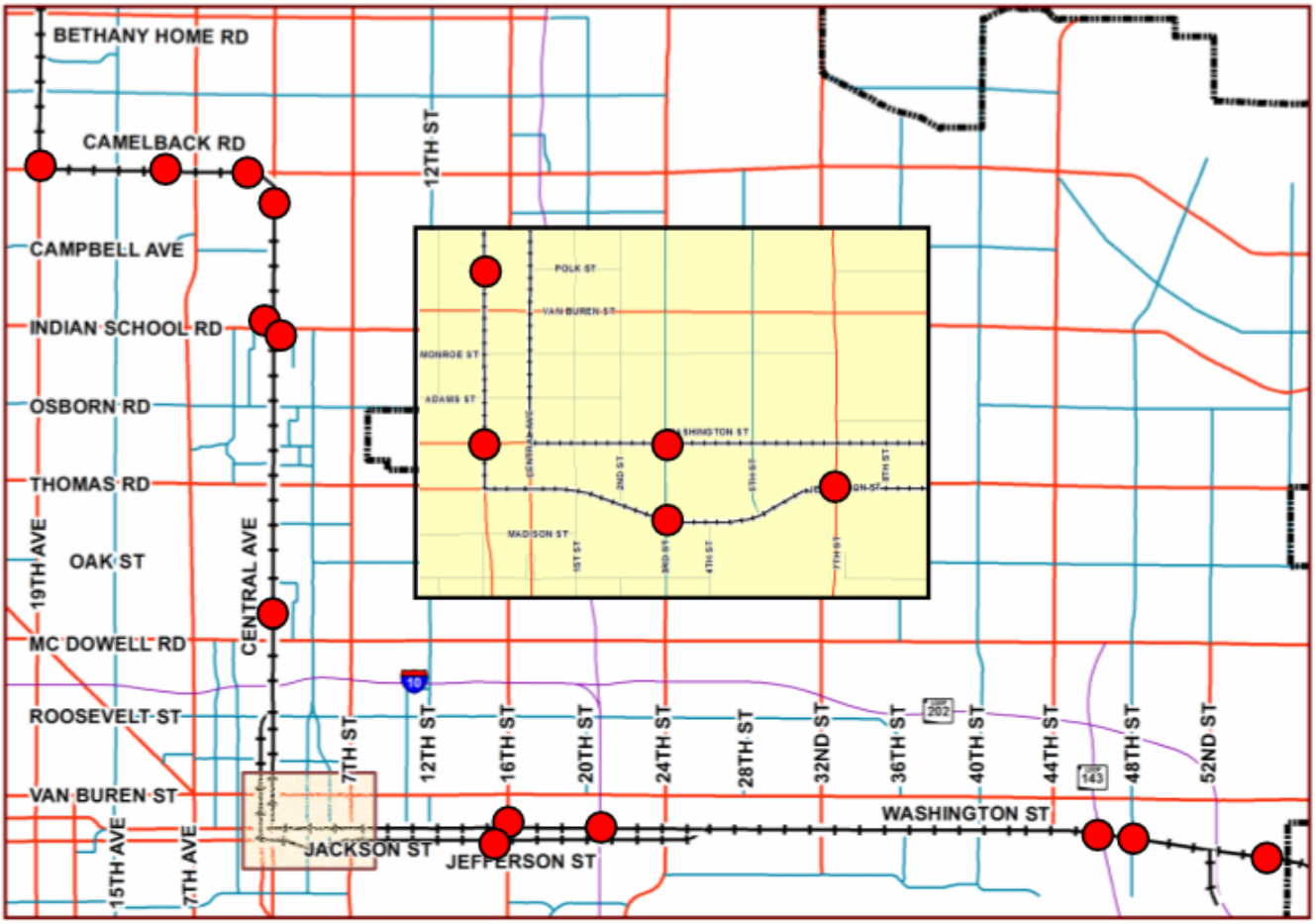


Motorcyclist Injury Severity and Helmet Usage



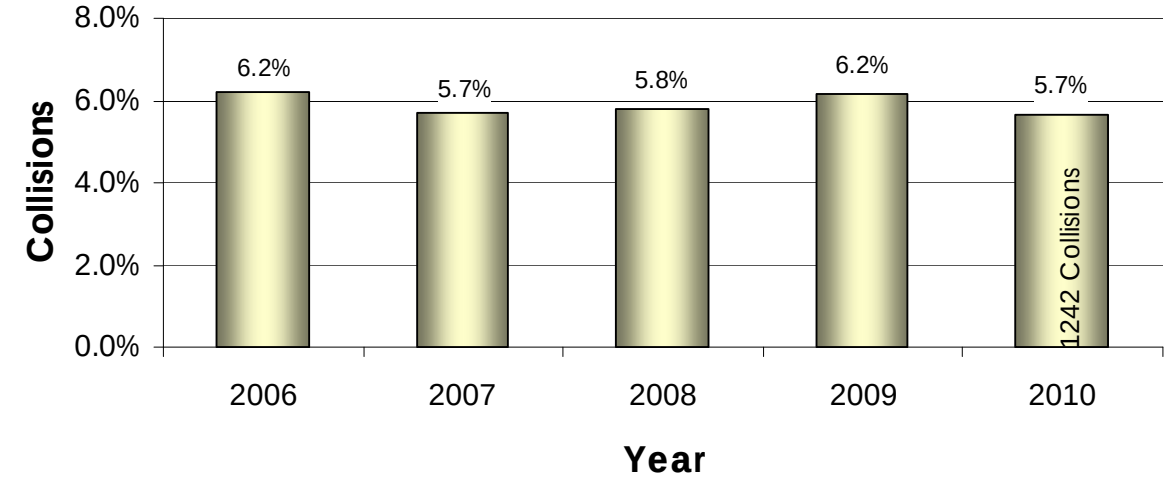
Light Rail Transit (LRT) Related Collisions 2010

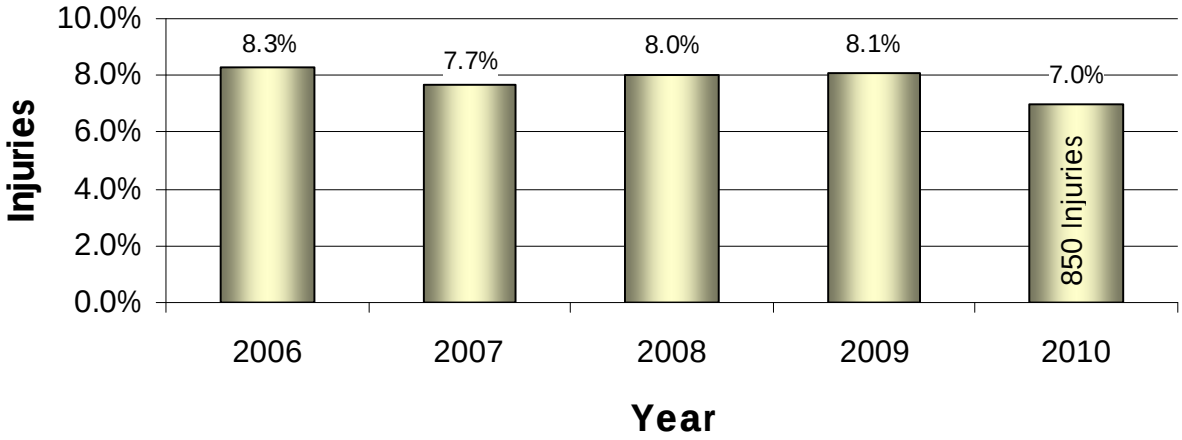


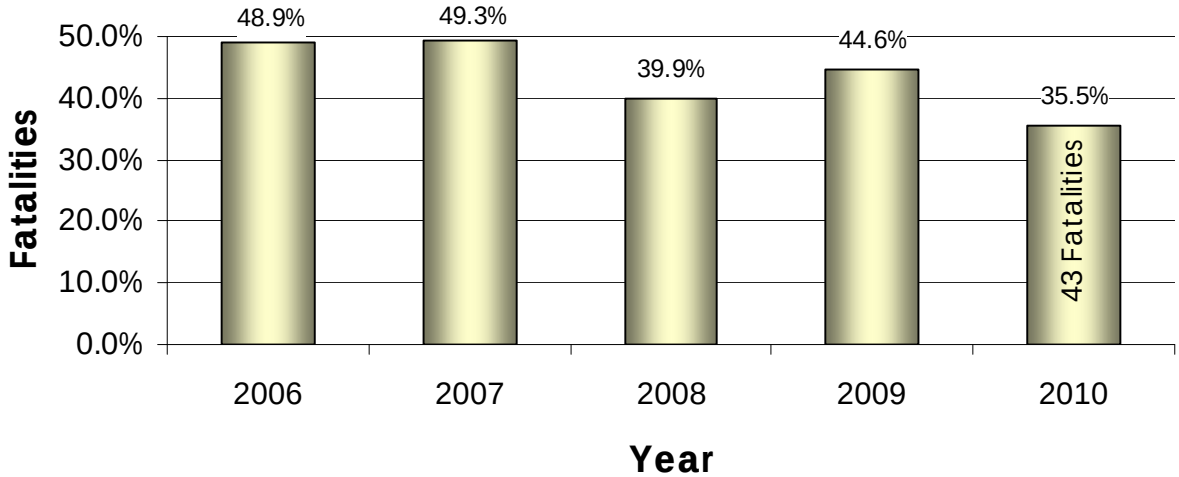


*There were 4 crashes not included in the database that were included in these statistics.

Alcohol-Related Collisions, Injuries, and Fatalities 2006–2010

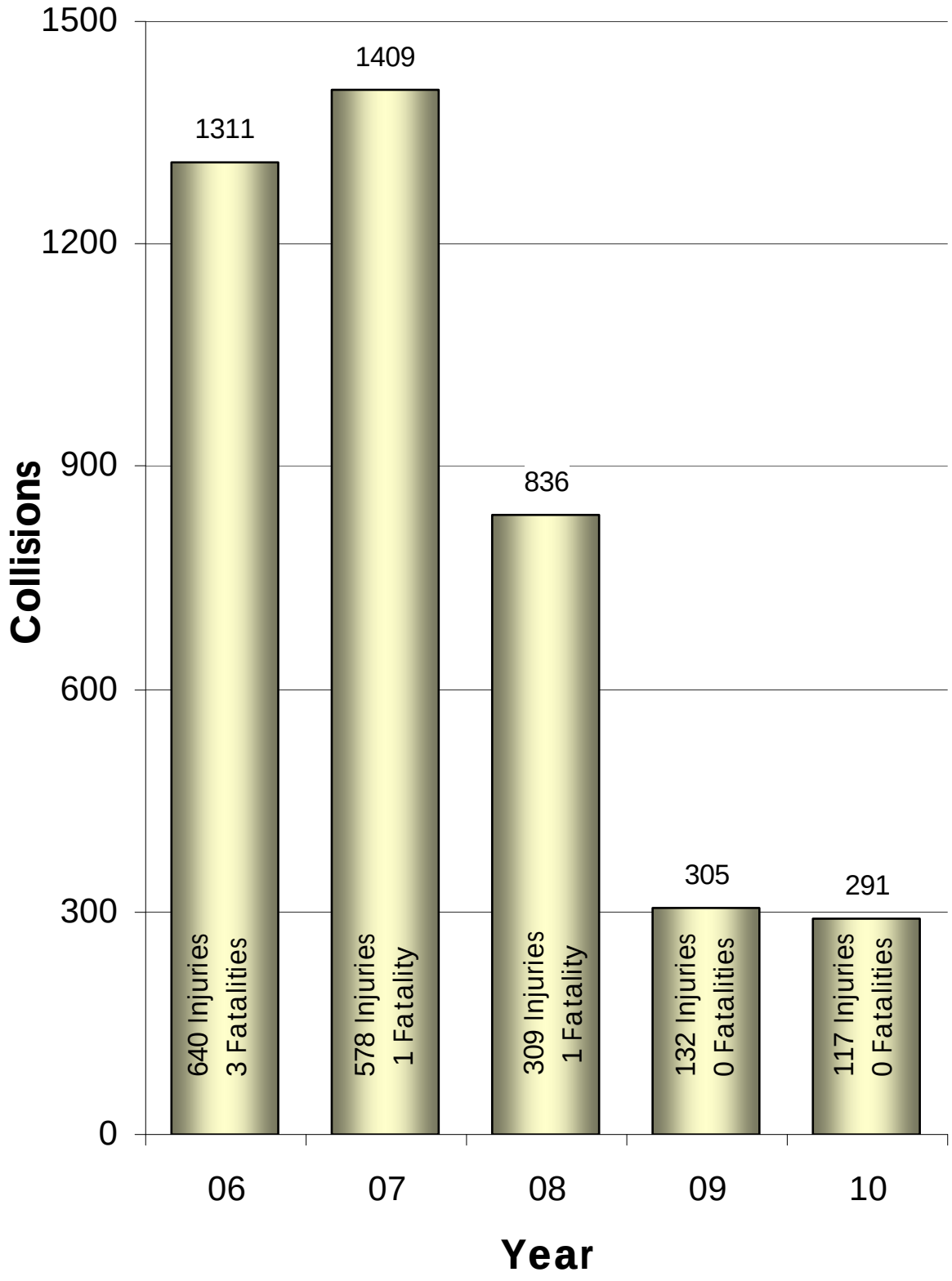




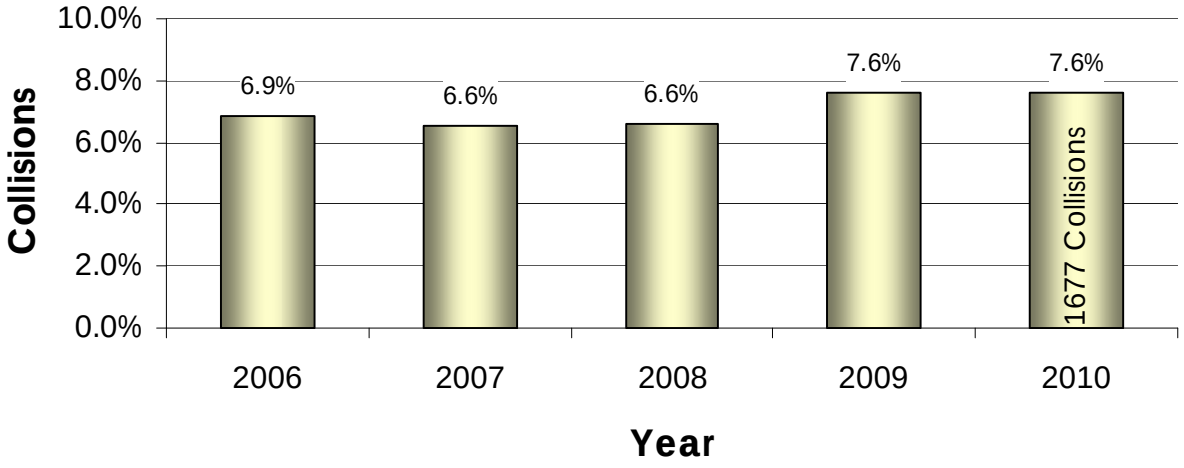


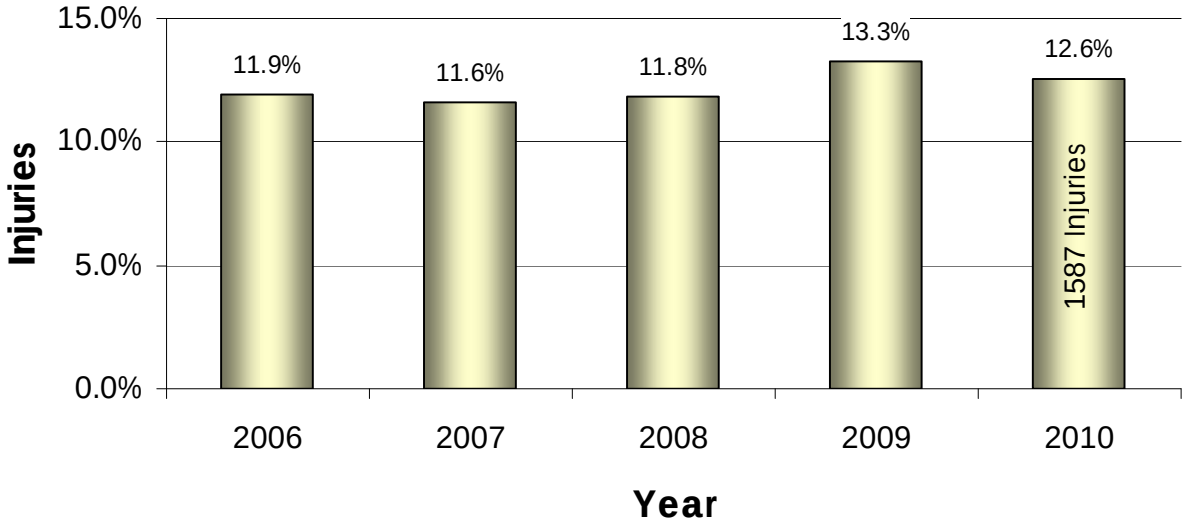
These graphs represent the percentage of alcohol-related crashes, injuries and fatalities in relation to the total number of crashes, injuries and fatalities. "Alcohol-Related" may not mean that the operator was legally drunk.

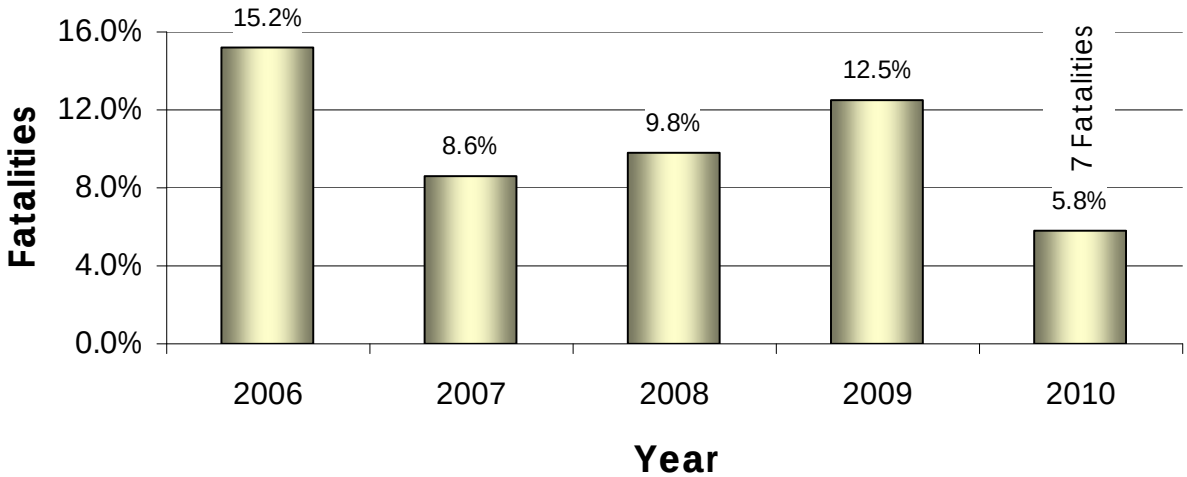
Work Zone Collisions, Injuries, and Fatalities 2006 – 2010



Run-Red Collisions, Injuries, and Fatalities 2006 – 2010*

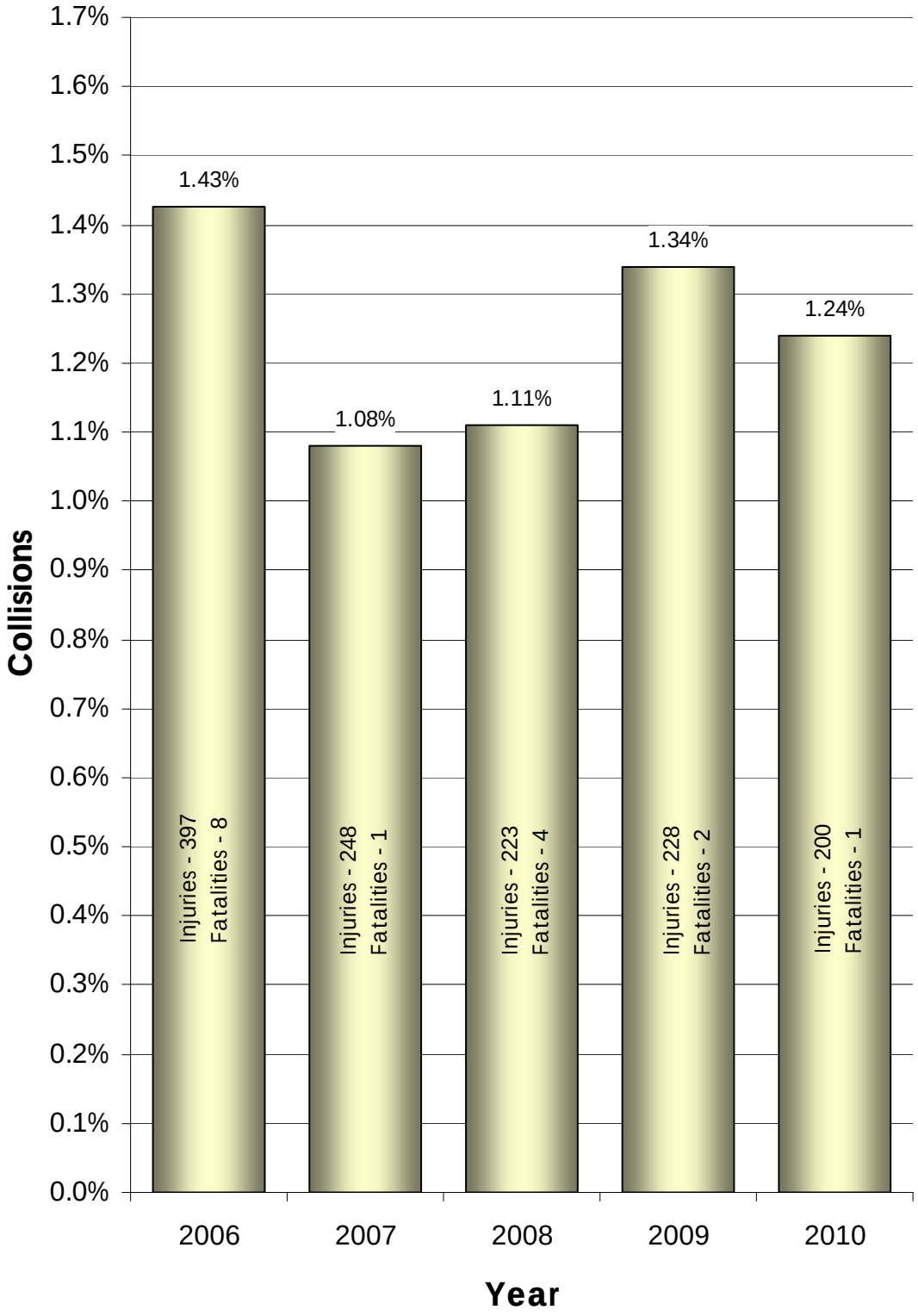






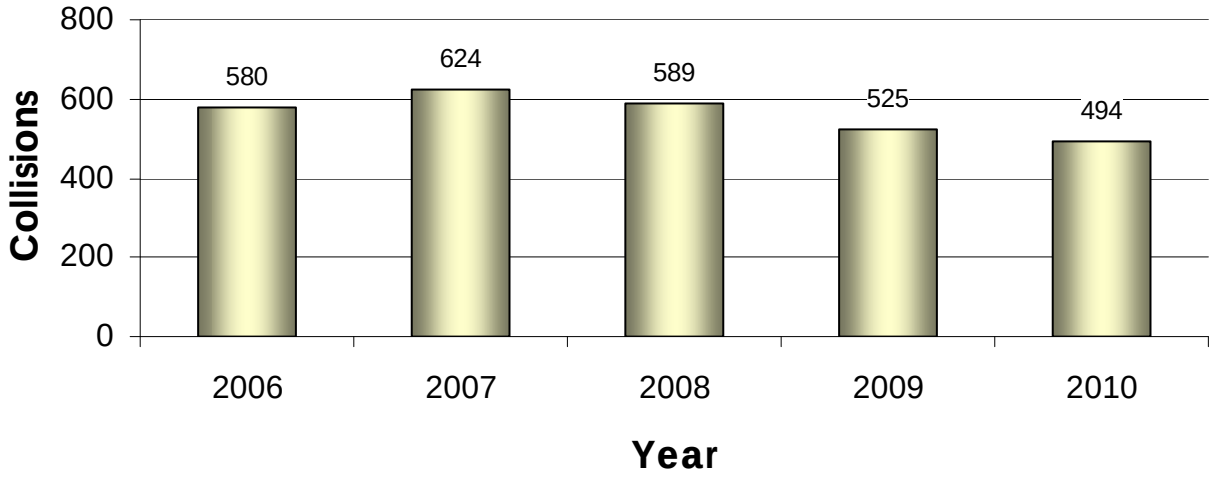
* Only collisions in which a motorist committed the violation are included.

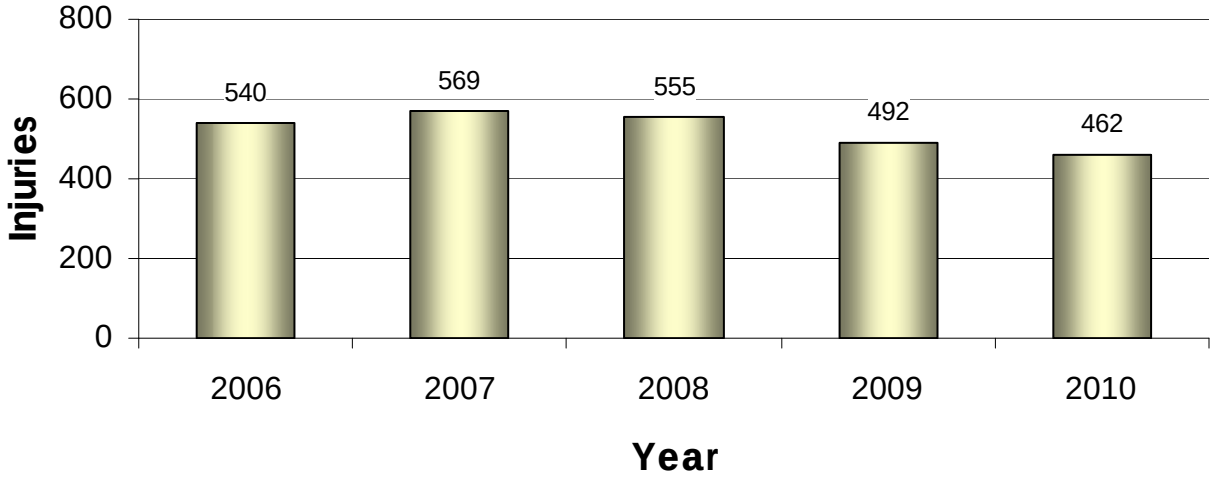
Run STOP Sign Collisions, Injuries, and Fatalities 2006 – 2010*

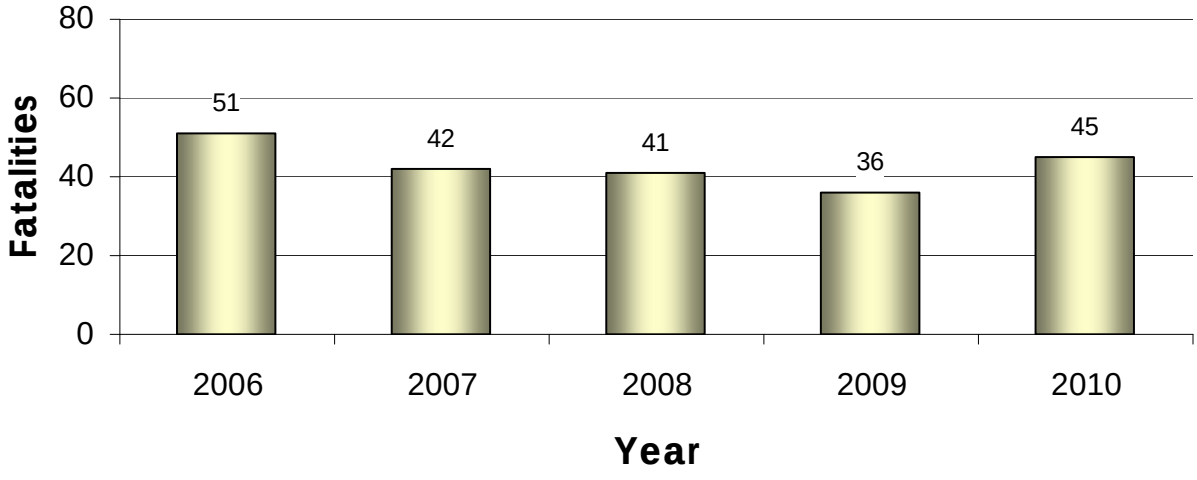


*Only collisions in which a motorist committed the violation are included. These graphs represent the percentage of "Run STOP Sign" crashes in relation to the total number of crashes.

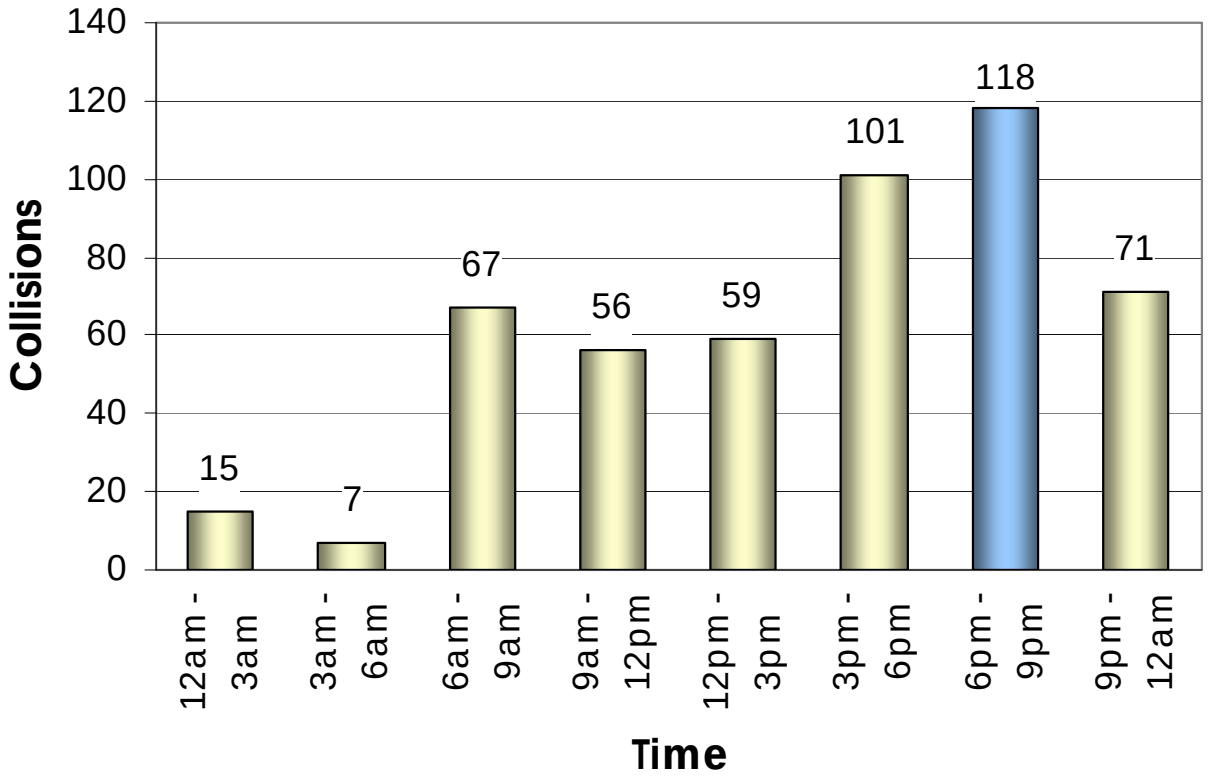
Pedestrian Collisions, Injuries, and Fatalities 2006 - 2010

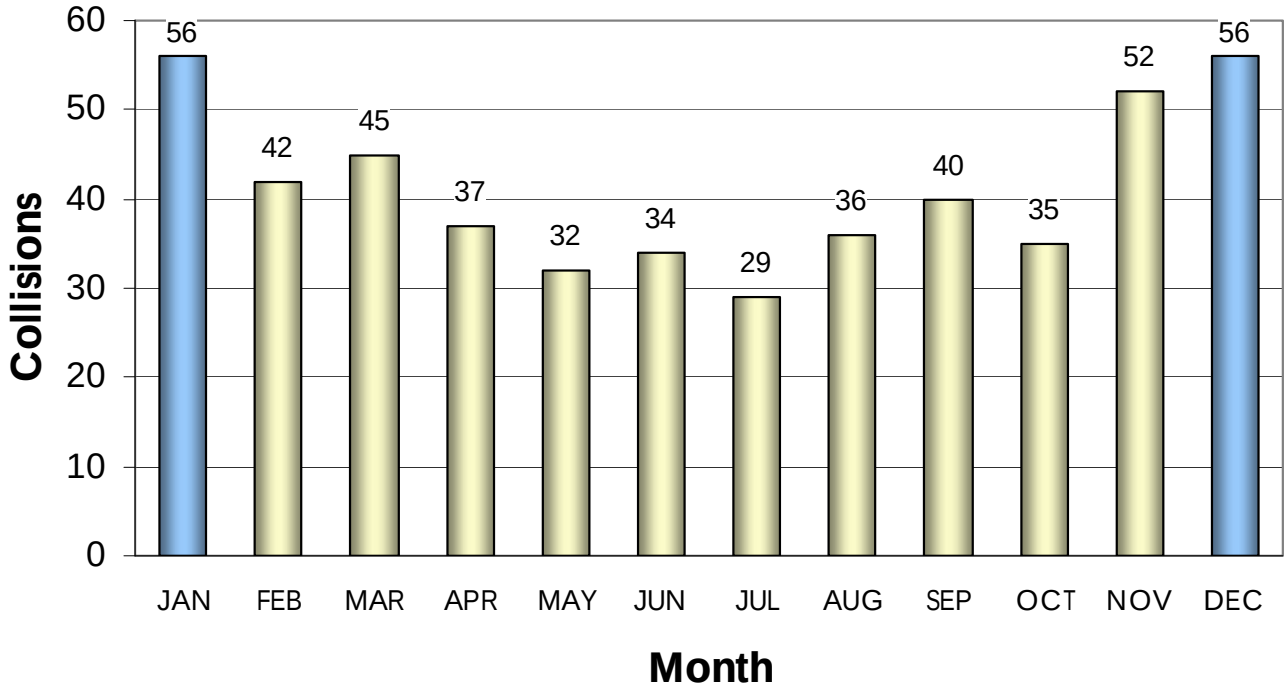






Pedestrian Collisions by Hour and Month

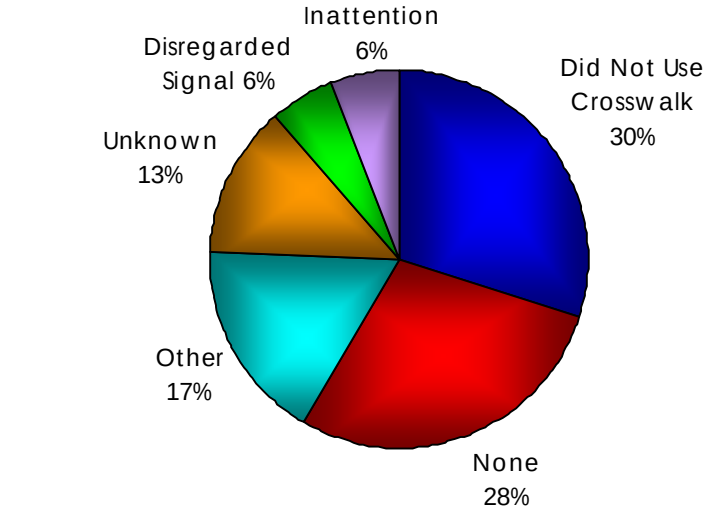




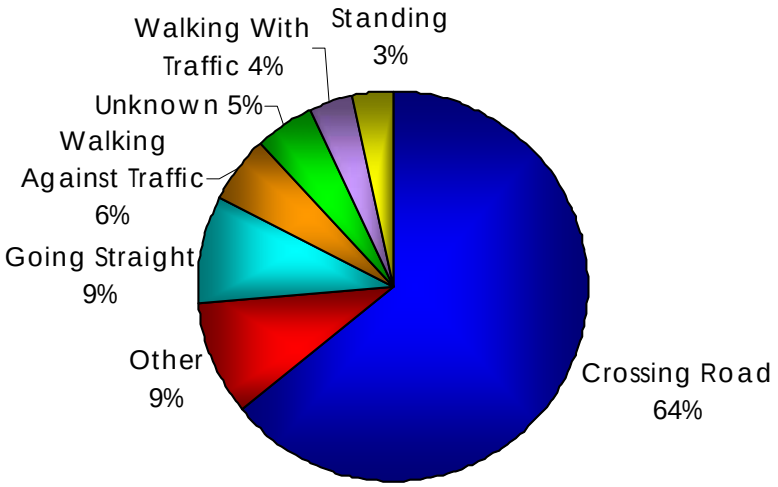
Note: Blue Column is Highest Month in 2010

Factors in Pedestrian Collisions: Pedestrian Behavior

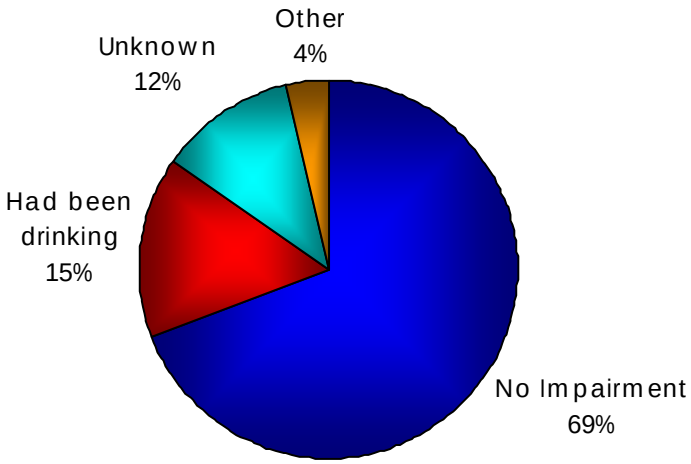
Violation by Pedestrian



Action of Pedestrian



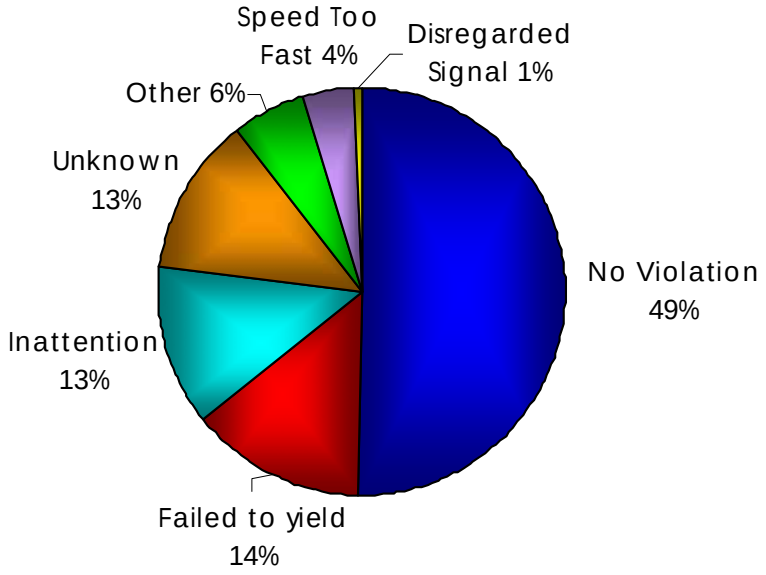
Physical Condition of Pedestrian



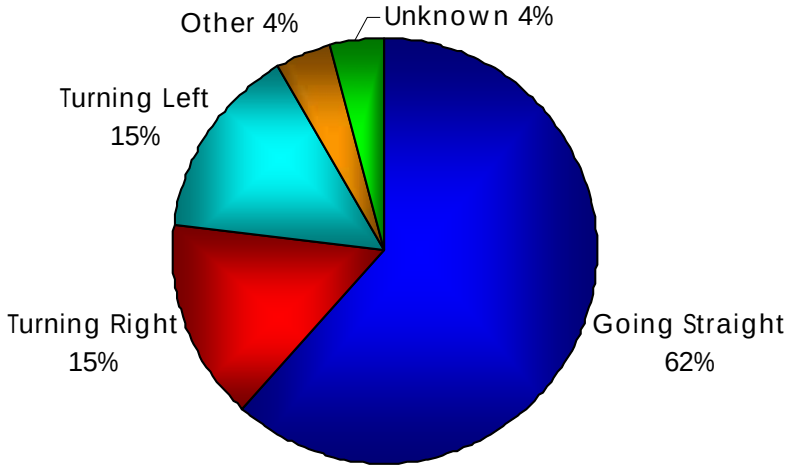
* The "Other" option is reported on a collision report form when none of the options accurately describe the attributes of the collision.

Factors in Pedestrian Collisions: Motorist Behavior

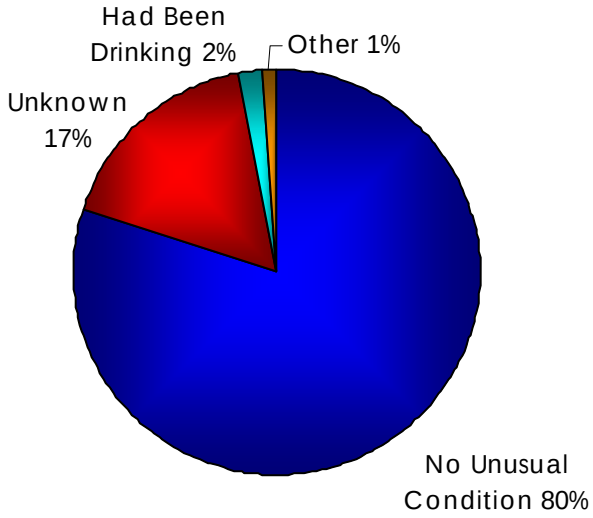
Violation by Motorist



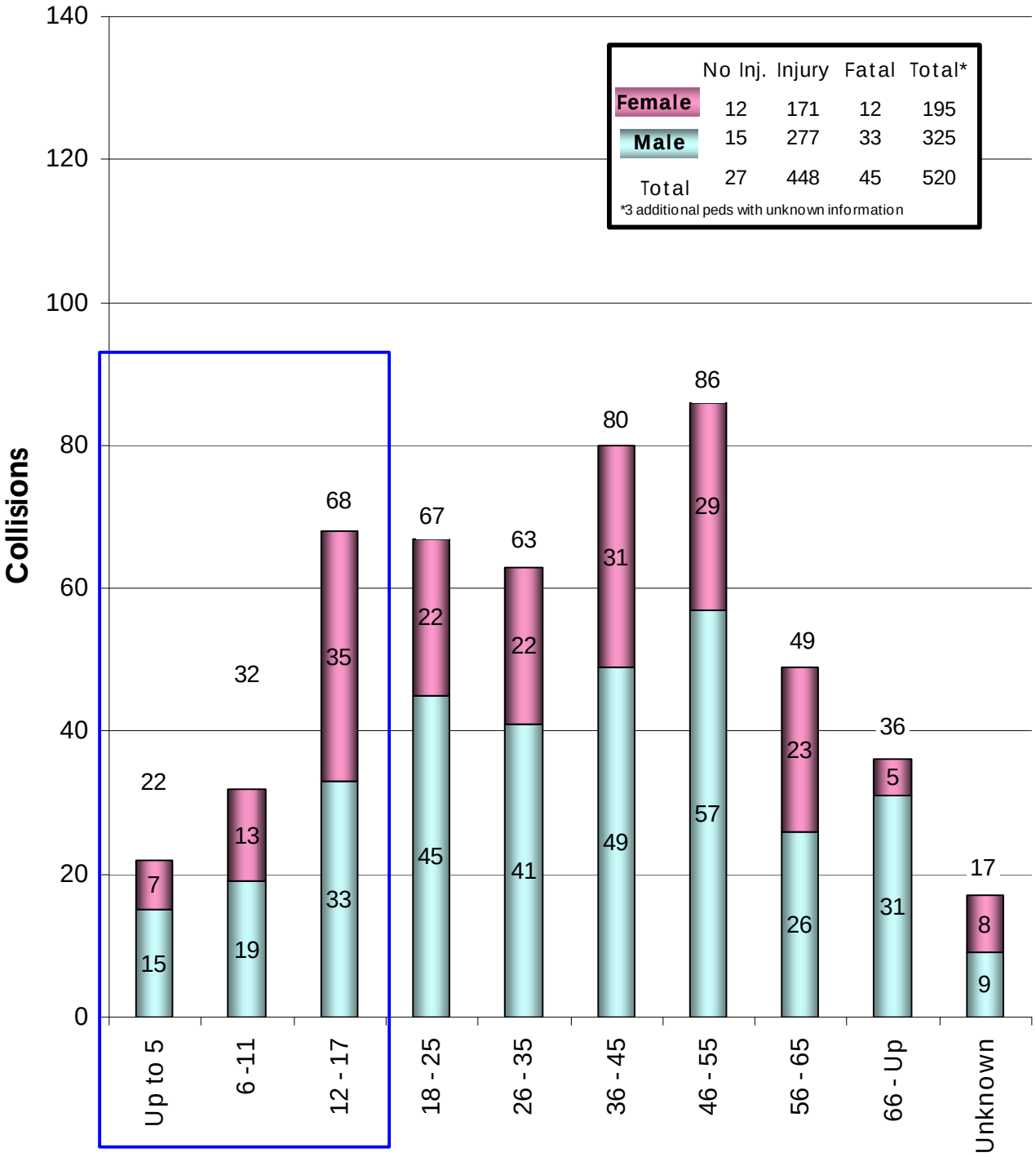
Action of Motorist



Physical Condition of Motorist

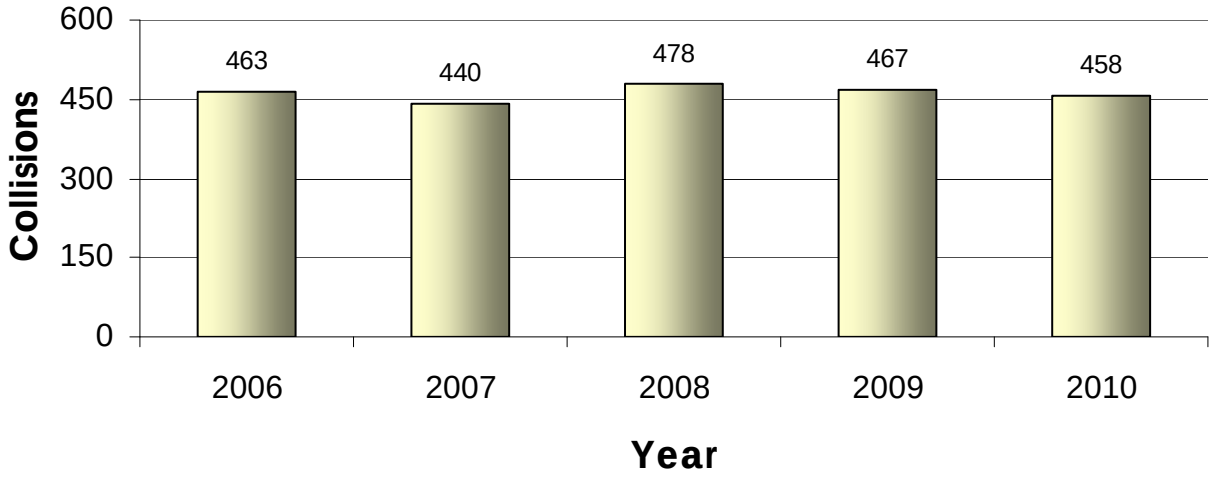


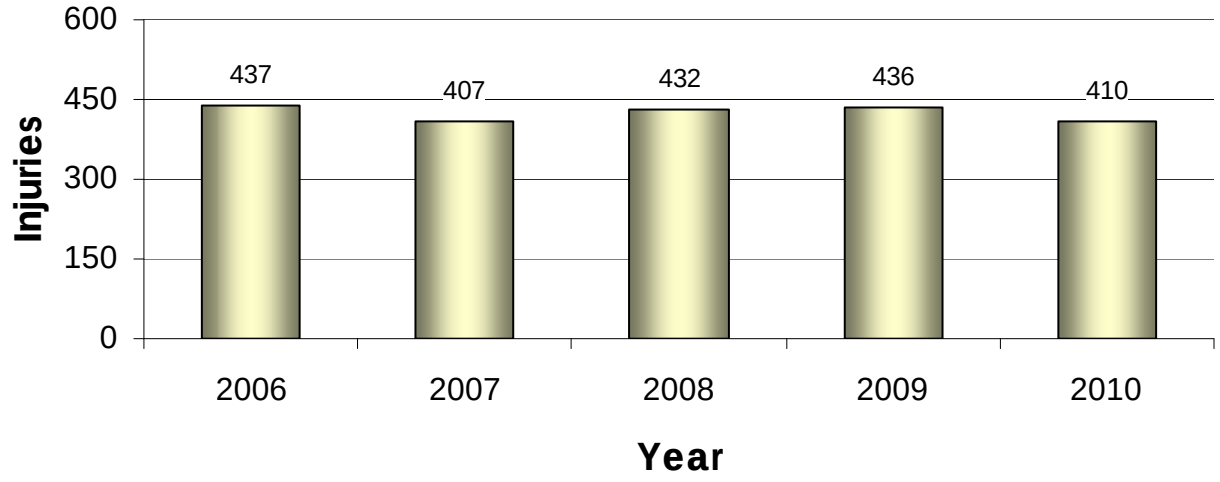
Age and Gender of Pedestrians in Collisions

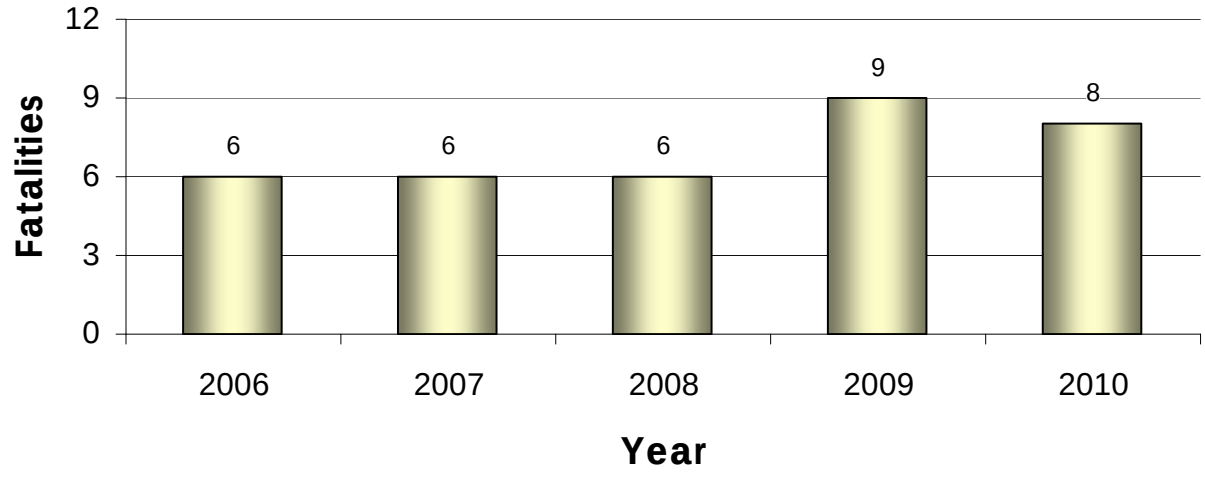


Children below 18 years of age account for 23% of all pedestrian crash victims

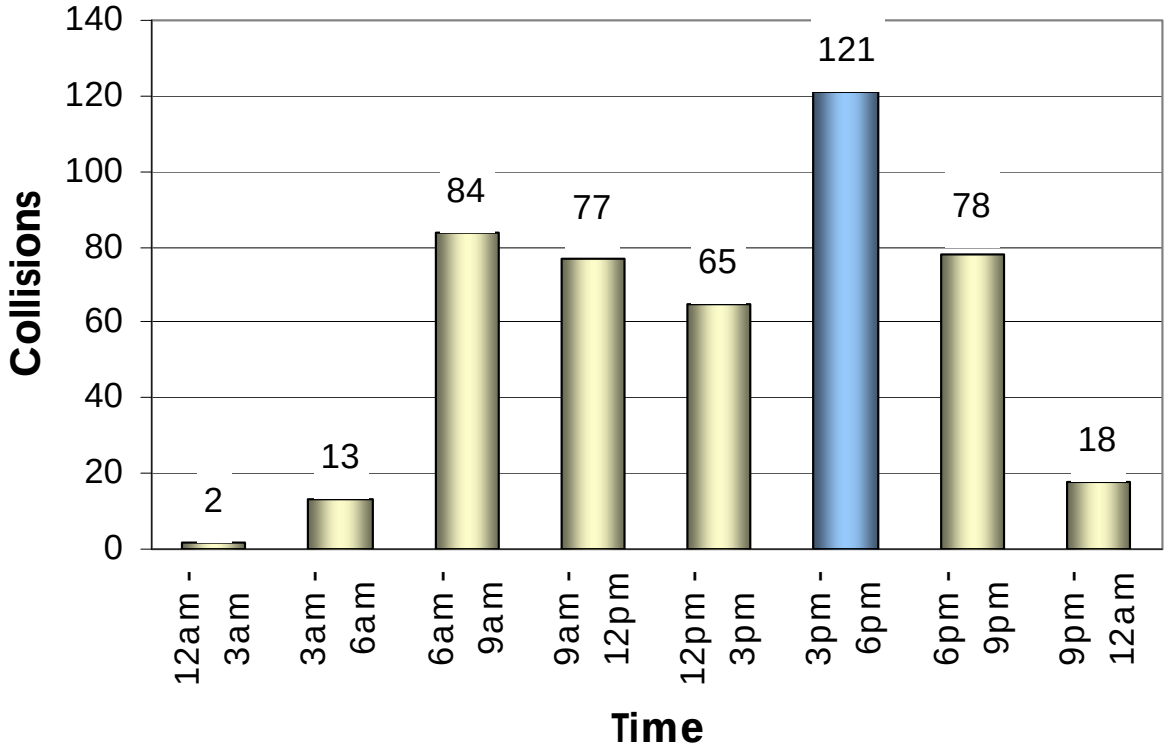
Bicycle Collisions, Injuries, and Fatalities 2006 - 2010

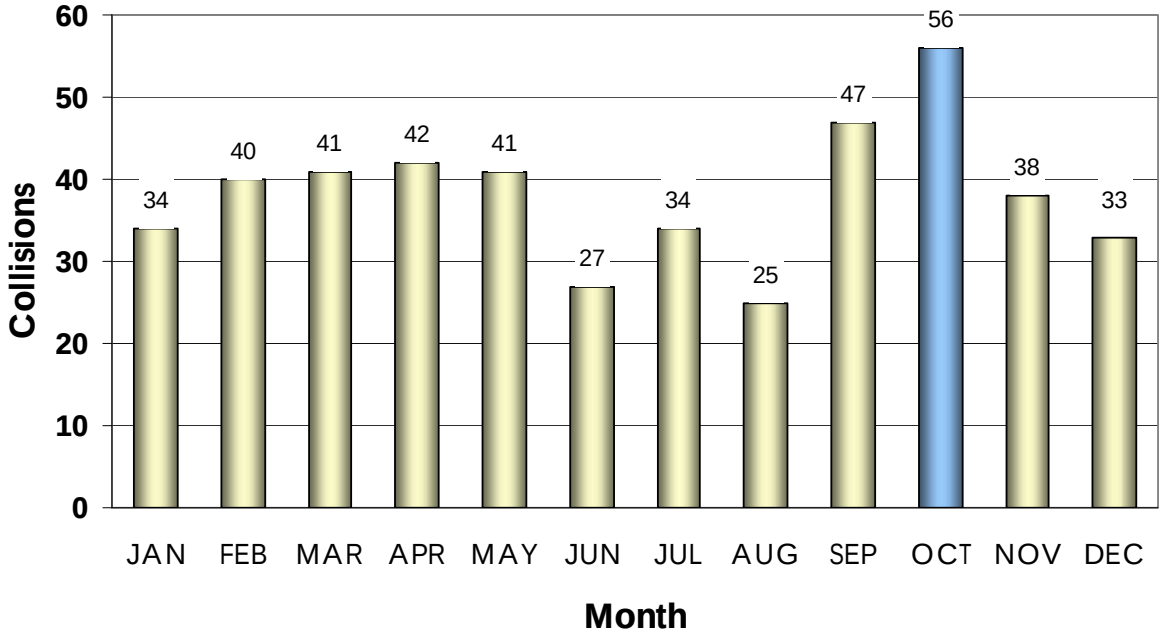






Bicycle Collisions by Hour and Month

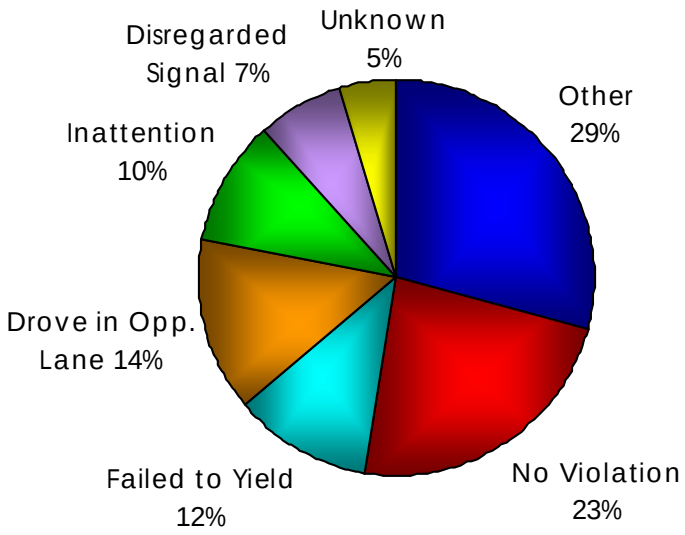




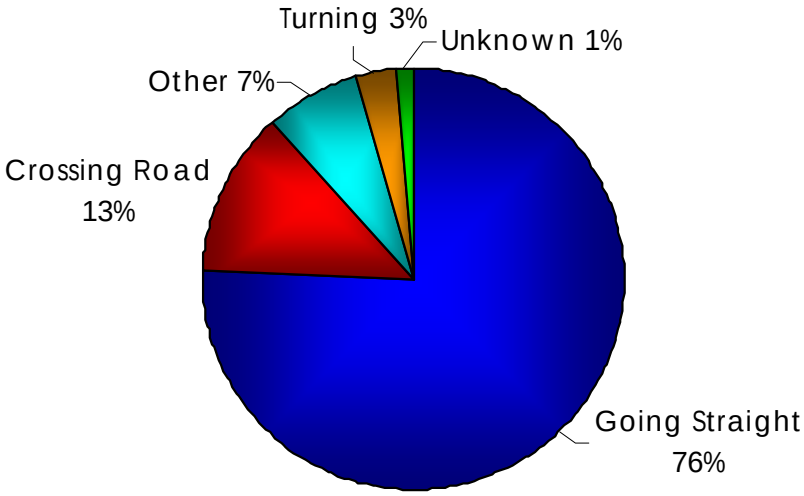
Note: Blue Column is Highest Month in 2010

Factors in Bicycle Collisions: Bicyclist Behavior

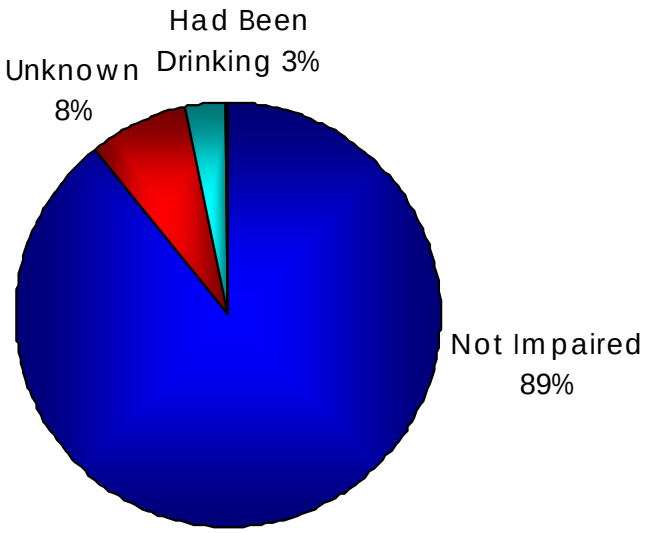
Violation by Bicyclist



Action of Bicyclist

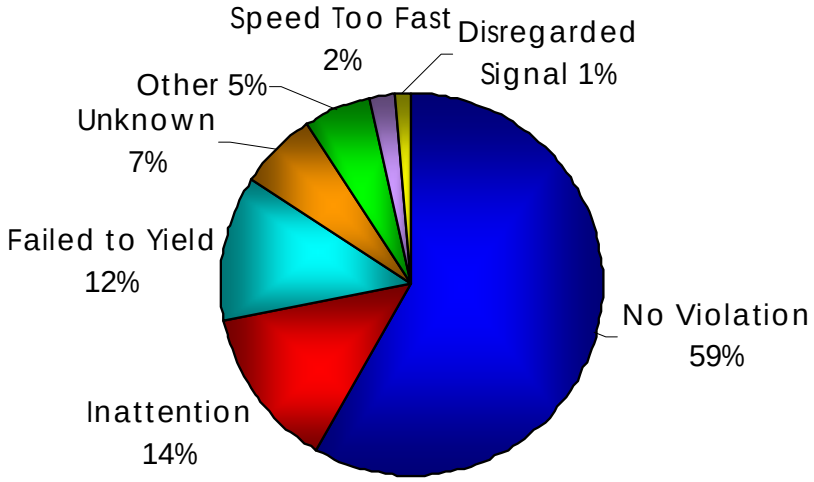


Physical Condition of Bicyclist

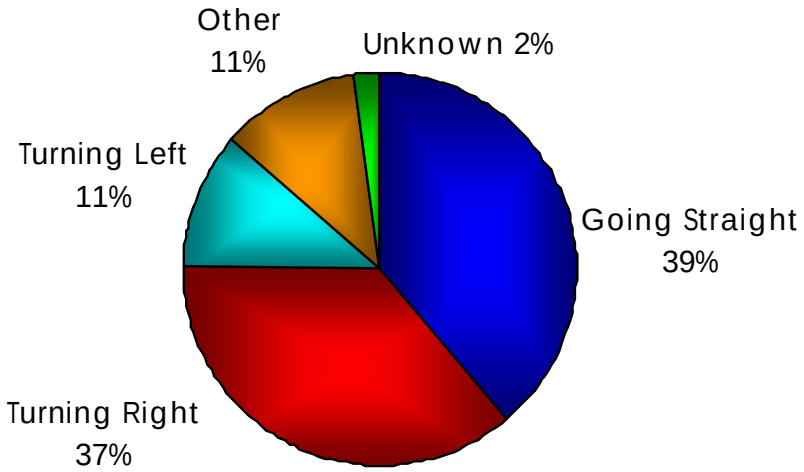


Factors in Bicycle Collisions: Motorist Behavior

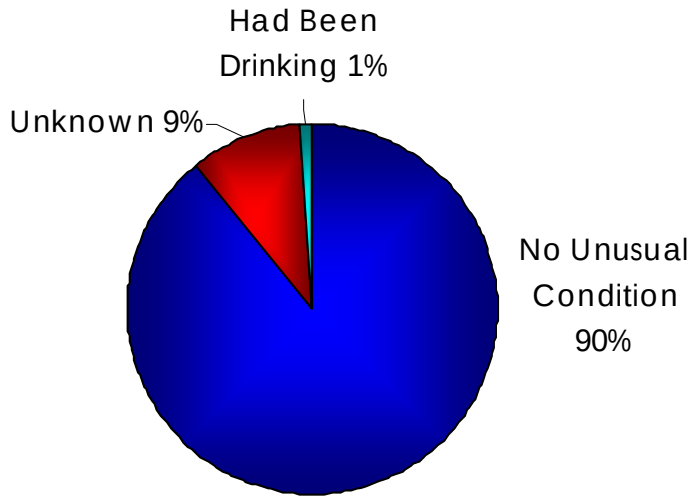
Violation by Motorist



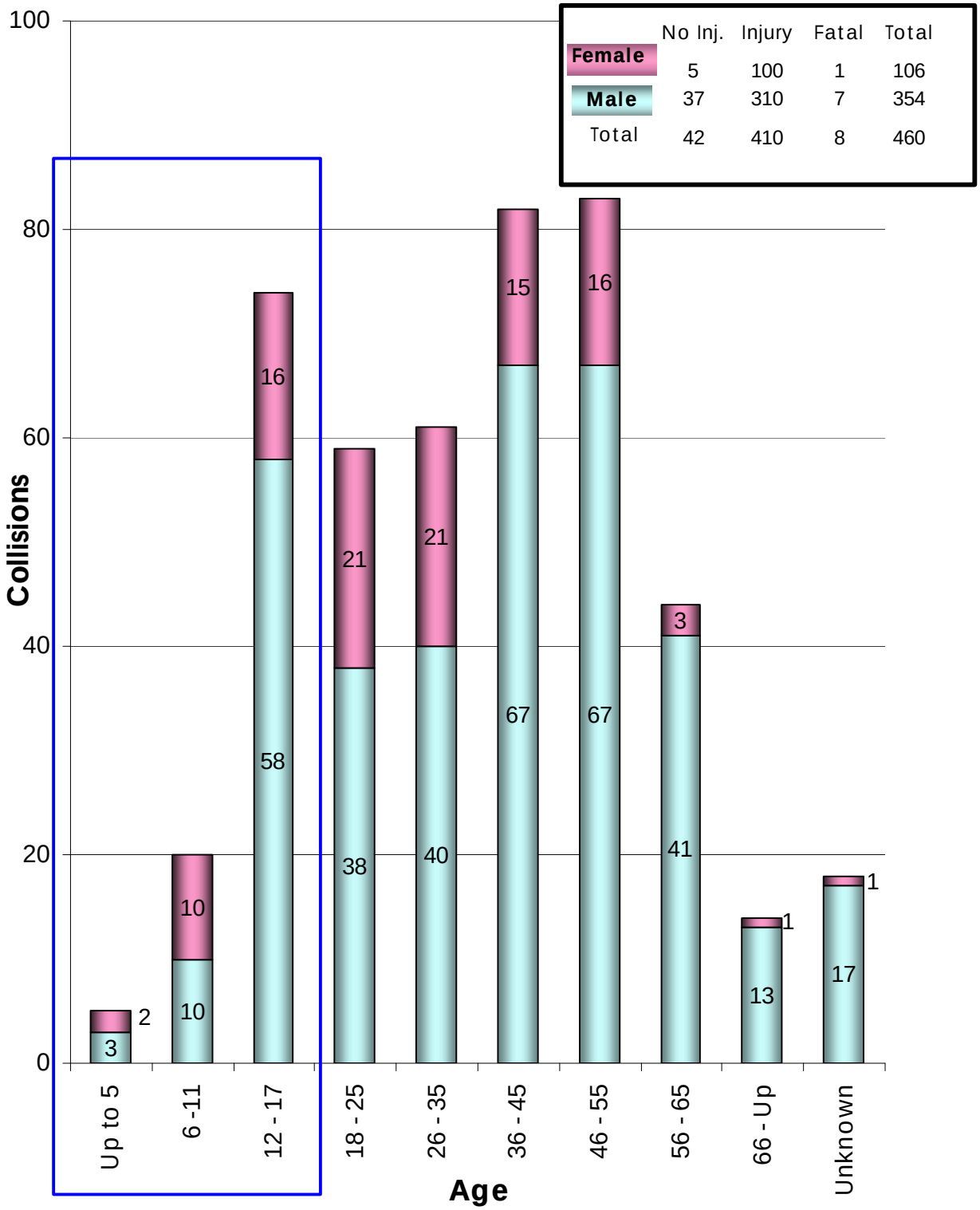
Action of Motorist



Physical Condition of Motorist



Age and Gender of Bicyclists in Collisions

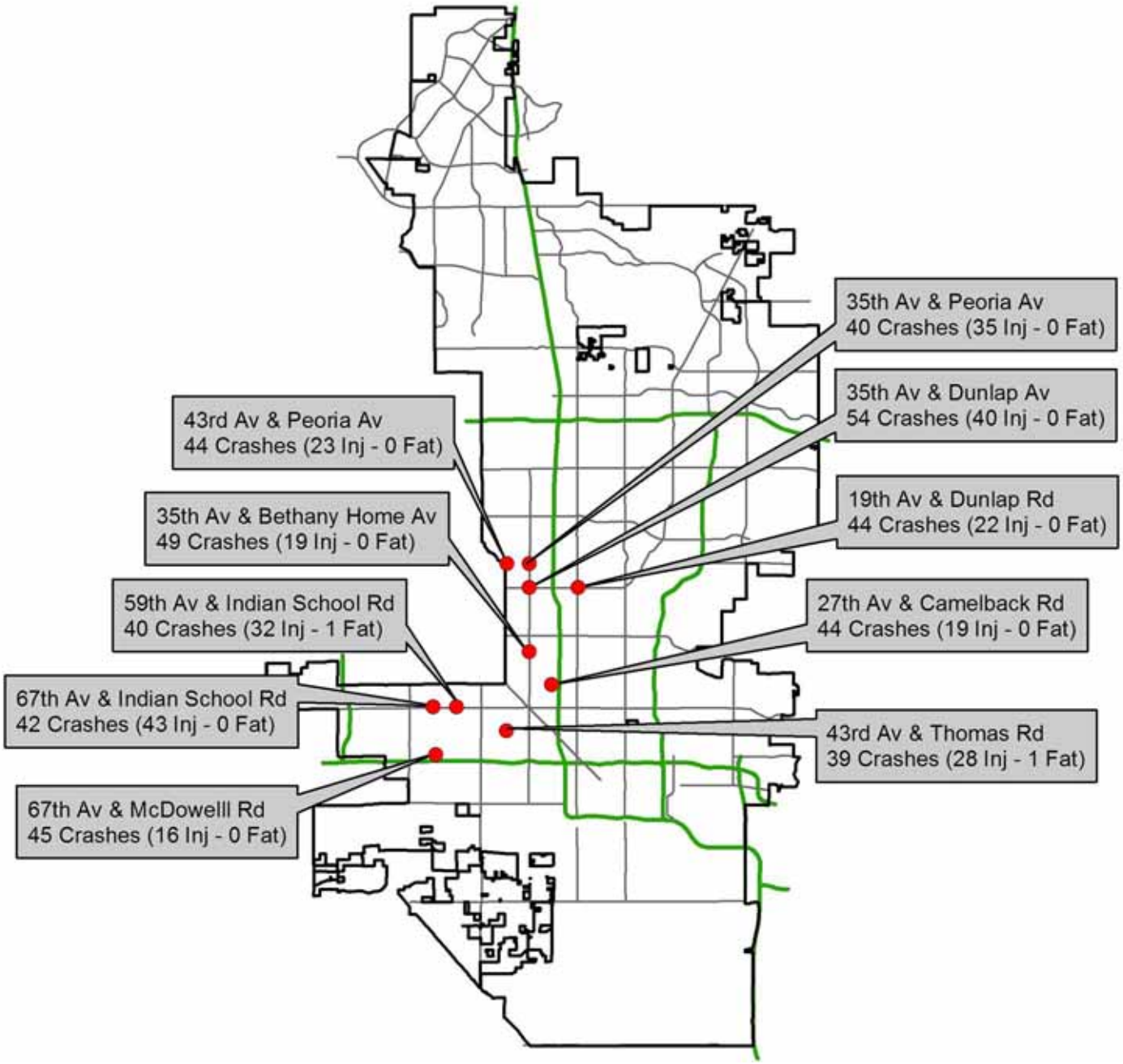


Children below 18 years of age account for almost 22% of all bicycle crash victims

Locations with the Highest Number of Collisions

Typically, intersections with the highest traffic volumes experience the most collisions. Since the listing does not account for severity, the diagram should not be interpreted to represent intersections of most concern from a safety standpoint.

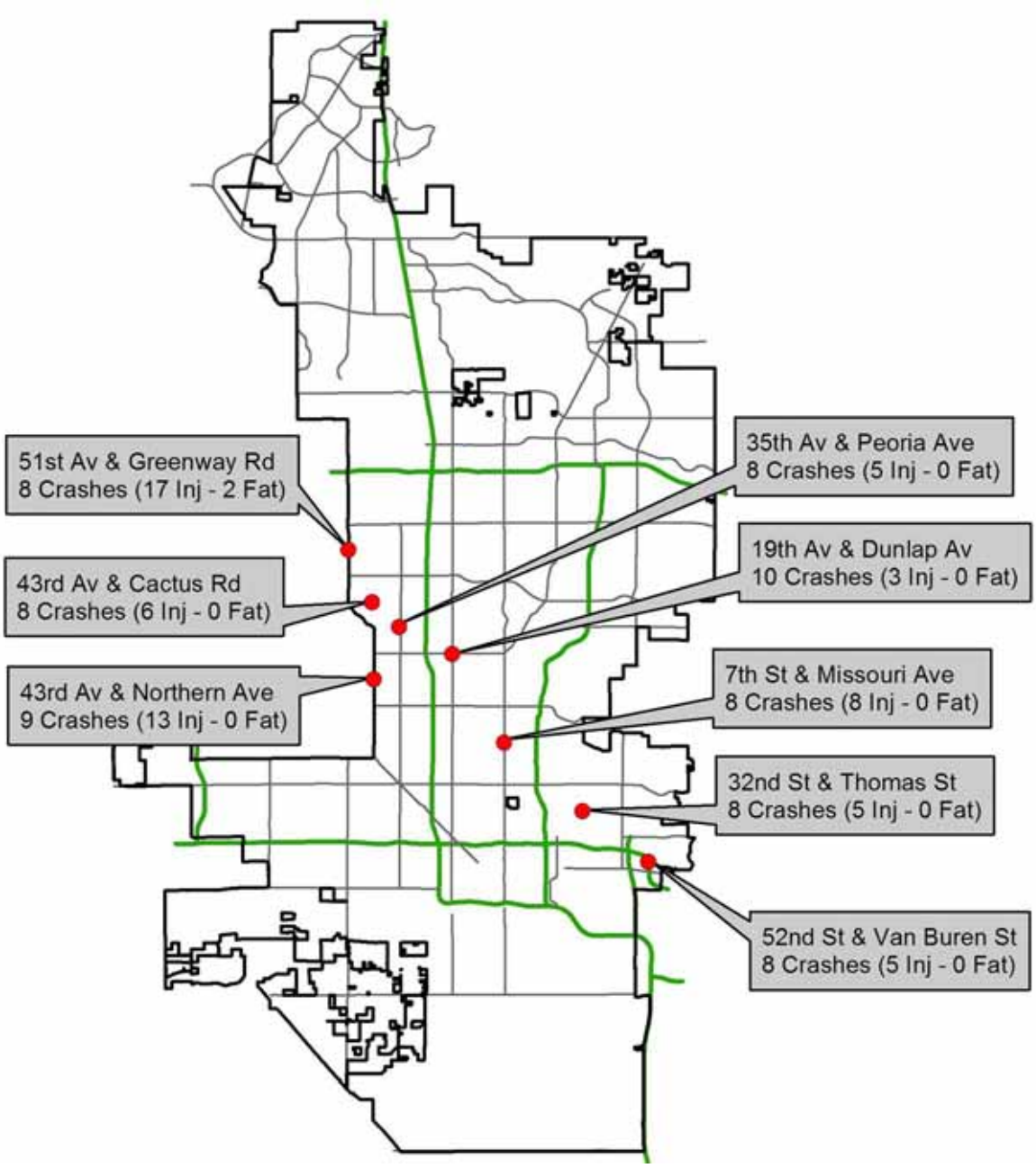
Criteria used to select these locations include all collisions in 2010 within 150 feet of an intersection.



Locations with the Highest Number of Run-Red Collisions

Typically, intersections with the highest traffic volumes experience the most collisions. Since the listing does not account for severity, the diagram should not be interpreted to represent intersections of most concern from a safety standpoint.

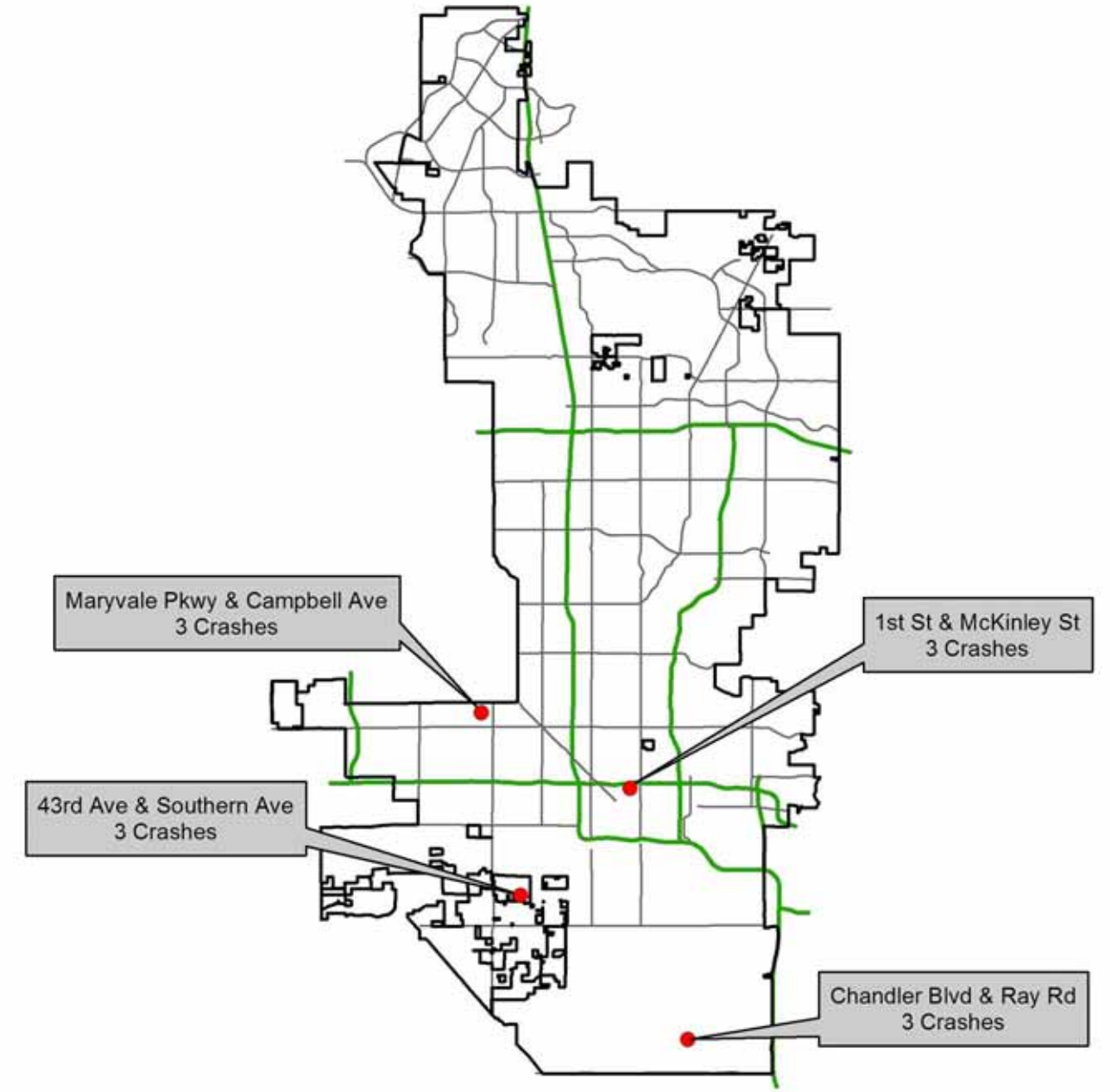
Criteria used to select these locations include all collisions in 2010 occurring within 150 feet of the intersection that resulted from a "Disregarded Traffic Signal" violation by a motorist.



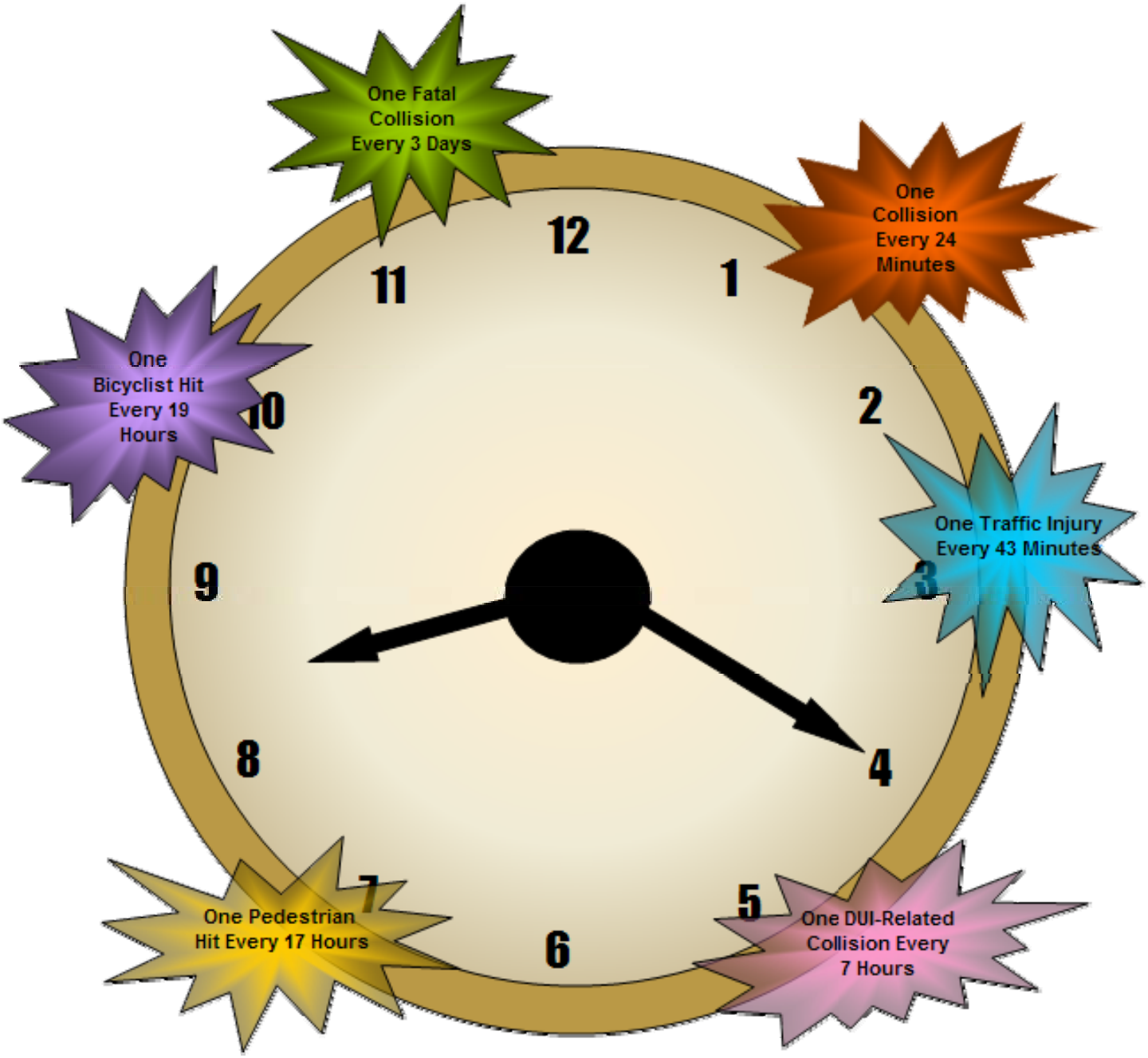
Locations with the Highest Number of Ran STOP Sign Collisions

Typically, intersections with the highest traffic volumes experience the most collisions. Since the listing does not account for severity, the diagram should not be interpreted to represent intersections of most concern from a safety standpoint.

Criteria used to select these locations include all collisions in 2010 occurring within 150 feet of the intersection that resulted from a "Ran STOP Sign" violation by a motorist.



Phoenix Traffic Clock



Additional Information

The 2010 Phoenix Traffic Collision Summary is the annual report of traffic collision statistics from the City of Phoenix. Data used to produce this document were obtained from the Arizona Department of Transportation's Safety Data Mart (SDM). SDM includes all collisions investigated under Arizona Revised Statutes 28-667 involving motor vehicles on city streets that involve injuries, fatalities, damage to private property in excess of $1000, or issuance of a citation. Collisions that occur on the freeway system or private property are not included in this report. SDM data may not match statistics reported by other sources.

For more information or to receive this document in an alternative format, please call the Street Transportation Department at 602-262-6284 or TTY 602-256-6284.